notebook

May 3, 2026

1 Problem Statement: Analysis of Student Dropout Determinants in Kenyan Higher Education

1.1 1. Background & Context

Student dropout from higher education institutions is a critical challenge in Kenya, leading to significant individual, social, and economic costs. These include wasted educational resources, reduced skilled workforce, increased unemployment, and lower lifetime earnings for the individuals who drop out. Understanding the multifaceted factors that contribute to this phenomenon is essential for developing effective, data-driven retention strategies. This dataset captures information on 118,000+ students, providing a rich opportunity to explore the complex interplay of academic, financial, demographic, and engagement-related factors.

1.2 2. Problem Identification

The core problem is the unexplained and unmitigated student dropout phenomenon in the Kenyan higher education context. Specifically, we investigate: - What is the primary driver of dropout: academic preparedness, current performance, or financial constraints? - How do financial factors interact with academic factors to influence dropout risk? - Does the impact vary significantly by demographics, programme of study, or engagement levels? - Can a predictive model identify students at high risk of dropping out early?

1.3 3. Objectives

1. **Descriptive Analysis** — Profile the student population and compare characteristics of dropouts vs. persisters.
2. **Exploratory Analysis** — Investigate relationships between independent variables and dropout status.
3. **Predictive Modelling** — Develop and evaluate a stacking classifier to predict dropout risk.
4. **Actionable Insights** — Translate findings into recommendations for administrators and policymakers.

1.4 1. Imports

[2]:
```
# Core libraries
import pandas as pd
import numpy as np
from datetime import datetime
```

```python
# Visualisation
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.cm as cm

# Statistics
from scipy.stats import chi2_contingency, ttest_ind

# Preprocessing
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer

# Models
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import StackingClassifier
!pip install imblearn
from imblearn.ensemble import BalancedRandomForestClassifier
from xgboost import XGBClassifier

# Model selection
from sklearn.model_selection import (
    train_test_split, RandomizedSearchCV
)

# Metrics
from sklearn.metrics import (
    accuracy_score, classification_report, confusion_matrix,
    ConfusionMatrixDisplay, roc_auc_score, f1_score, recall_score
)

# Pipeline
from sklearn.pipeline import Pipeline

# Warnings
import warnings
warnings.filterwarnings('ignore')
```

```
Requirement already satisfied: imblearn in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (0.0)
Requirement already satisfied: imbalanced-learn in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from imblearn)
(0.14.1)
Requirement already satisfied: numpy<3,>=1.25.2 in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from
imbalanced-learn->imblearn) (2.4.4)
Requirement already satisfied: scipy<2,>=1.11.4 in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from
imbalanced-learn->imblearn) (1.17.1)
```

```
Requirement already satisfied: scikit-learn<2,>=1.4.2 in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from
imbalanced-learn->imblearn) (1.8.0)
Requirement already satisfied: sklearn-compat<0.2,>=0.1.5 in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from
imbalanced-learn->imblearn) (0.1.5)
Requirement already satisfied: joblib<2,>=1.2.0 in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from
imbalanced-learn->imblearn) (1.5.3)
Requirement already satisfied: threadpoolctl<4,>=2.0.0 in
C:\Users\Administrator\anaconda3\envs\project\Lib\site-packages (from
imbalanced-learn->imblearn) (3.6.0)
```

1.5 2. Data Loading

[3]:
```
School_df = pd.read_csv("student.csv")
School_df.head()
```

[3]:
```
  Gender   Birthdate  PHYSICALLYCHALLENGED  County  Loan_serial_number  \
0   Male  13/03/1996                     0  MIGORI          2510374563
1   Male  08/07/1982                     0     NaN          2510057518
2   Male  18/02/1993                     0     NaN          2510233585
3   Male  18/03/1986                     0     NaN          2510031694
4   Male  31/10/1987                     0  NAKURU          2510438107

   ScholarshipApplied InstitutionCode                InstitutionName  \
0                   0             KPU  TECHNICAL UNIVERSITY OF KENYA
1                   0              CA            Catholic University
2                   0             MTK         Mount Kenya University
3                   0          KAFUCO     Kaimosi Friends University
4                   0             UBG              Bugema University

  COURSECODE  ProgramCost  …                       Category applicanttype  \
0       EEEI     270000.0  …   PublicUniversityGovtSponsored             1
1        BED     160000.0  …  PrivateUniversitySelfSponsored             1
2       BBIT     111000.0  …  PrivateUniversitySelfSponsored             1
3        HNR     275400.0  …   PublicUniversitySelfSponsored             1
4        BHT          0.0  …  PrivateUniversitySelfSponsored             1

            LoanStatus Unnamed: 21  Mother_Age Father_Age  Father_educ_level  \
0  Partially Disbursed         NaN          69         64            Primary
1  Partially Disbursed         NaN          63         65            Diploma
2  Partially Disbursed         NaN          54         57            Primary
3            Allocated         NaN          51         76            Diploma
4  Partially Disbursed         NaN          58         64          Secondary

   Mother_educ_level Highest_educ_level  Drop_out
```

```
0            Primary            Primary         0
1          Secondary            Diploma         1
2          Secondary          Secondary         0
3            Primary            Diploma         0
4          Secondary          Secondary         0

[5 rows x 28 columns]
```

[4]:
```python
print(School_df.shape)
School_df.info()
```

```
(118259, 28)
<class 'pandas.DataFrame'>
RangeIndex: 118259 entries, 0 to 118258
Data columns (total 28 columns):
 #   Column                Non-Null Count   Dtype
---  ------                --------------   -----
 0   Gender                62659 non-null   str
 1   Birthdate             118259 non-null  str
 2   PHYSICALLYCHALLENGED  118259 non-null  int64
 3   County                118235 non-null  str
 4   Loan_serial_number    118259 non-null  int64
 5   ScholarshipApplied    118259 non-null  int64
 6   InstitutionCode       118259 non-null  str
 7   InstitutionName       118259 non-null  str
 8   COURSECODE            117794 non-null  str
 9   ProgramCost           117794 non-null  float64
 10  CourseCategory        114059 non-null  str
 11  Adm No                118233 non-null  str
 12  ExamYear              118259 non-null  int64
 13  INDEXNUMBER           118258 non-null  object
 14  TotalLoanAllocated    118259 non-null  float64
 15  LoanproductCode       118259 non-null  str
 16  BursaryAllocated      118259 non-null  int64
 17  BURSARYBATCHNUMBER    0 non-null       float64
 18  Category              118259 non-null  str
 19  applicanttype         118259 non-null  int64
 20  LoanStatus            118256 non-null  str
 21  Unnamed: 21           0 non-null       float64
 22  Mother_Age            118259 non-null  int64
 23  Father_Age            118259 non-null  int64
 24  Father_educ_level     101442 non-null  str
 25  Mother_educ_level     97691 non-null   str
 26  Highest_educ_level    104711 non-null  str
 27  Drop_out              118259 non-null  int64
dtypes: float64(4), int64(9), object(1), str(14)
memory usage: 25.3+ MB
```

[5]:
```
School_df.isna().sum()
```

[5]:
```
Gender                   55600
Birthdate                    0
PHYSICALLYCHALLENGED         0
County                      24
Loan_serial_number           0
ScholarshipApplied           0
InstitutionCode              0
InstitutionName              0
COURSECODE                 465
ProgramCost                465
CourseCategory            4200
Adm No                      26
ExamYear                     0
INDEXNUMBER                  1
TotalLoanAllocated           0
LoanproductCode              0
BursaryAllocated             0
BURSARYBATCHNUMBER      118259
Category                     0
applicanttype                0
LoanStatus                   3
Unnamed: 21             118259
Mother_Age                   0
Father_Age                   0
Father_educ_level        16817
Mother_educ_level        20568
Highest_educ_level       13548
Drop_out                     0
dtype: int64
```

1.6 3. Data Cleaning

1.6.1 3.1 Drop Irrelevant Columns

[6]:
```
columns_to_drop = [
    'BURSARYBATCHNUMBER', 'Unnamed: 21', 'PHYSICALLYCHALLENGED',
    'Loan_serial_number', 'InstitutionCode', 'Adm No',
    'COURSECODE', 'INDEXNUMBER', 'BursaryAllocated', 'applicanttype'
]
School_df = School_df.drop(columns=columns_to_drop, errors='ignore')
School_df.columns
```

[6]:
```
Index(['Gender', 'Birthdate', 'County', 'ScholarshipApplied',
       'InstitutionName', 'ProgramCost', 'CourseCategory', 'ExamYear',
       'TotalLoanAllocated', 'LoanproductCode', 'Category', 'LoanStatus',
       'Mother_Age', 'Father_Age', 'Father_educ_level', 'Mother_educ_level',
```

```
       'Highest_educ_level', 'Drop_out'],
      dtype='str')
```

1.6.2 3.2 Drop Rows with Missing Values in Key Columns

[7]:
```
School_df = School_df.dropna(subset=[
    "Gender", "County", "ProgramCost", "CourseCategory",
    "Father_educ_level", "Mother_educ_level", "LoanStatus", "Highest_educ_level"
])
School_df.isna().sum()
```

[7]:
```
Gender                0
Birthdate             0
County                0
ScholarshipApplied    0
InstitutionName       0
ProgramCost           0
CourseCategory        0
ExamYear              0
TotalLoanAllocated    0
LoanproductCode       0
Category              0
LoanStatus            0
Mother_Age            0
Father_Age            0
Father_educ_level     0
Mother_educ_level     0
Highest_educ_level    0
Drop_out              0
dtype: int64
```

1.6.3 3.3 Remove Unsupported Loan Product Codes

[8]:
```
School_df = School_df[
    ~School_df['LoanproductCode'].isin(['DL6', 'VC'])
]
School_df['LoanproductCode'].value_counts()
```

[8]:
```
LoanproductCode
UG    47652
Name: count, dtype: int64
```

1.6.4 3.4 Remove Duplicates

[9]:
```
print("Duplicates before:", School_df.duplicated().sum())
School_df = School_df.drop_duplicates()
print("Duplicates after:", School_df.duplicated().sum())
```

```
Duplicates before: 6
Duplicates after: 0
```

1.7 4. Feature Engineering

1.7.1 Assumptions

- Age 16+ is eligible to join an undergraduate programme.

1.7.2 4.1 Create Age Column

[10]:
```
# Convert Birthdate to datetime and derive age
School_df["Birthdate"] = pd.to_datetime(School_df["Birthdate"], errors="coerce")
today = datetime.today()
School_df["age"] = School_df["Birthdate"].apply(
    lambda x: today.year - x.year - ((today.month, today.day) < (x.month, x.
 ↪day))
    if pd.notnull(x) else None
)
```

1.7.3 4.2 Extract University Type and Sponsorship from Category

[11]:
```
School_df[['UniversityType', 'Sponsored']] = School_df['Category'].str.extract(
    r'^(Public|Private).*?(GovtSponsored|SelfSponsored)$'
)
School_df[['UniversityType', 'Sponsored']].value_counts()
```

[11]:
```
UniversityType  Sponsored
Public          GovtSponsored    41972
Private         SelfSponsored     3699
Public          SelfSponsored     1132
Private         GovtSponsored      843
Name: count, dtype: int64
```

1.7.4 4.3 Create Age Groups

[12]:
```
bins   = [0, 18, 21, 25]
labels = ['<17', '18-21', '22-25']
School_df['age_group'] = pd.cut(School_df['age'], bins=bins, labels=labels)
School_df['age_group'].value_counts()
```

[12]:
```
age_group
18-21    35738
```

```
<17        5935
22-25      5324
Name: count, dtype: int64
```

1.7.5 4.4 Export Cleaned Dataset

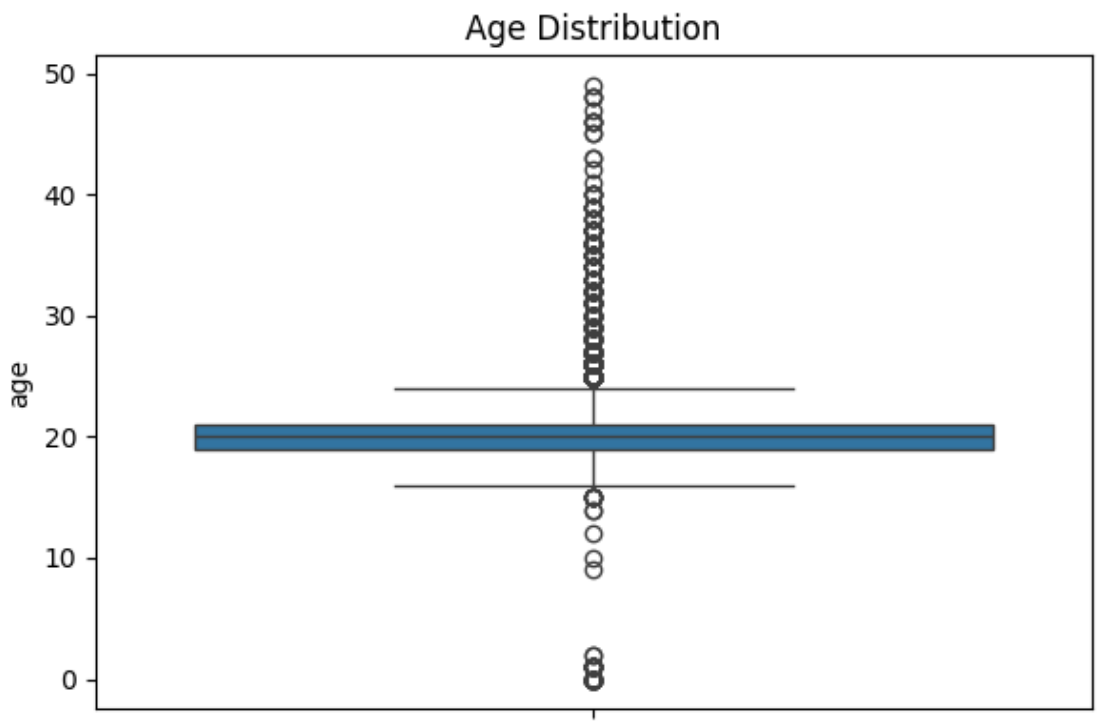
[13]:
```
School_df.to_csv('cleaned.csv', index=False)
print(f"Cleaned dataset saved: {School_df.shape[0]} rows, {School_df.shape[1]}␣
↪columns")
```

```
Cleaned dataset saved: 47646 rows, 22 columns
```

1.8 5. Exploratory Data Analysis

1.8.1 5.1 Age Distribution

[14]:
```
plt.figure(figsize=(6, 4))
sns.boxplot(y=School_df['age'])
plt.title('Age Distribution')
plt.tight_layout()
plt.show()
```



[15]:
```
# Filter to valid student age range to deal with
School_df = School_df[School_df['age'] >= 17]
```

```
School_df = School_df[(School_df['age'] >= 16) & (School_df['age'] <= 25)]
School_df['age'].describe()
```

[15]:
```
count    46883.000000
mean        19.801868
std          1.372580
min         17.000000
25%         19.000000
50%         20.000000
75%         20.000000
max         25.000000
Name: age, dtype: float64
```

[16]:
```
School_df.age.value_counts()
```

[16]:
```
age
19.0    17204
20.0    12156
21.0     6378
18.0     5148
22.0     3178
23.0     1339
17.0      673
24.0      556
25.0      251
Name: count, dtype: int64
```

1.8.2 5.2 School Drop Out Rate by County in Kenya

[17]:
```
# 1. Standardize to uppercase and strip whitespace
School_df['County'] = School_df['County'].astype(str).str.upper().str.strip()

# 2. Cleanup punctuation and spacing
# We replace slashes and hyphens with spaces, but we'll handle the apostrophe␣
↪carefully
School_df['County'] = School_df['County'].str.replace(r'[-/]', ' ', regex=True)
School_df['County'] = School_df['County'].str.replace(r'\s+', ' ', regex=True)

# 3. Specific fix for Murang'a and other duplicates
mapping = {
    'MURANG A': "MURANG'A",
    'MURANGA': "MURANG'A",
    'TRANSNZOIA': 'TRANS NZOIA'
}

School_df['County'] = School_df['County'].replace(mapping)
```

```python
# 4. Verification
unique_counties = sorted(School_df['County'].unique())
print(f"Unique County Count: {len(unique_counties)}")
print(unique_counties)
```

```
Unique County Count: 47
['BARINGO', 'BOMET', 'BUNGOMA', 'BUSIA', 'ELGEYO MARAKWET', 'EMBU', 'GARISSA',
'HOMA BAY', 'ISIOLO', 'KAJIADO', 'KAKAMEGA', 'KERICHO', 'KIAMBU', 'KILIFI',
'KIRINYAGA', 'KISII', 'KISUMU', 'KITUI', 'KWALE', 'LAIKIPIA', 'LAMU',
'MACHAKOS', 'MAKUENI', 'MANDERA', 'MARSABIT', 'MERU', 'MIGORI', 'MOMBASA',
"MURANG'A", 'NAIROBI', 'NAKURU', 'NANDI', 'NAROK', 'NYAMIRA', 'NYANDARUA',
'NYERI', 'SAMBURU', 'SIAYA', 'TAITA TAVETA', 'TANA RIVER', 'THARAKA NITHI',
'TRANS NZOIA', 'TURKANA', 'UASIN GISHU', 'VIHIGA', 'WAJIR', 'WEST POKOT']
```

[18]:
```python
School_df.County.unique()
```

[18]:
```
<StringArray>
[       'KAKAMEGA',         'KAJIADO',         'NAIROBI',          'KIAMBU',
           'KITUI',        'MACHAKOS',     'UASIN GISHU',         'MOMBASA',
          'MIGORI', 'ELGEYO MARAKWET',         'BUNGOMA',         'NYAMIRA',
          'KILIFI',           'BOMET',     'TRANS NZOIA',          'KISUMU',
        'HOMA BAY',            'MERU',   'THARAKA NITHI',           'KWALE',
        'MURANG'A',           'NYERI',           'BUSIA',          'NAKURU',
           'KISII',       'KIRINYAGA',           'NANDI',          'VIHIGA',
           'SIAYA',         'MAKUENI',       'NYANDARUA',    'TAITA TAVETA',
        'LAIKIPIA',         'KERICHO',            'LAMU',           'NAROK',
         'SAMBURU',         'BARINGO',            'EMBU',      'WEST POKOT',
         'GARISSA',        'MARSABIT',         'TURKANA',      'TANA RIVER',
          'ISIOLO',         'MANDERA',           'WAJIR']
Length: 47, dtype: str
```

[19]:
```python
# Create the summary table
dropout_table = School_df.groupby('County').agg(
    Total_Students = ('Drop_out', 'count'),
    Dropout_Count  = ('Drop_out', 'sum')
).reset_index()

# Calculate dropout percentage
dropout_table['Dropout_Percentage'] = (
    dropout_table['Dropout_Count'] / dropout_table['Total_Students'] * 100
).round(2)

# Sort by highest dropout percentage
dropout_table = dropout_table.sort_values('Dropout_Percentage', ascending=False)

# Reset index cleanly
dropout_table = dropout_table.reset_index(drop=True)
```

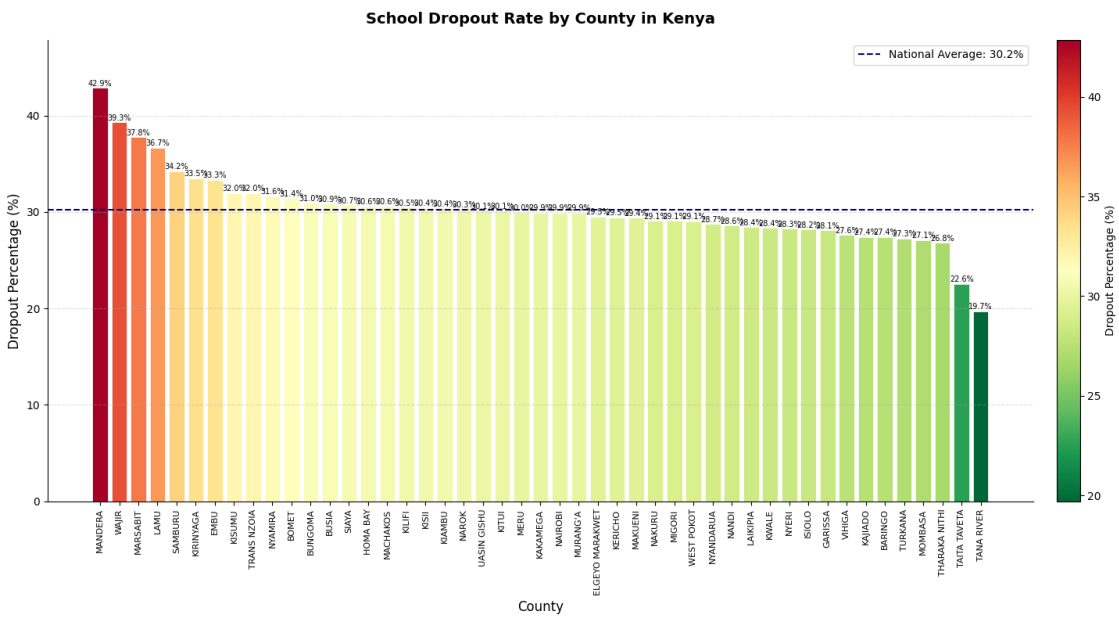
```r
print(dropout_table)
```

```
             County  Total_Students  Dropout_Count  Dropout_Percentage
0           MANDERA              21              9               42.86
1             WAJIR              28             11               39.29
2          MARSABIT              90             34               37.78
3              LAMU              60             22               36.67
4           SAMBURU              73             25               34.25
5         KIRINYAGA             373            125               33.51
6              EMBU             447            149               33.33
7            KISUMU            1858            594               31.97
8       TRANS NZOIA            1236            395               31.96
9           NYAMIRA            1725            546               31.65
10            BOMET            1387            436               31.43
11          BUNGOMA            2080            644               30.96
12            BUSIA            1295            400               30.89
13            SIAYA            1609            494               30.70
14         HOMA BAY            2913            891               30.59
15         MACHAKOS            1060            324               30.57
16           KILIFI             581            177               30.46
17            KISII            3366           1024               30.42
18           KIAMBU            1570            477               30.38
19            NAROK             706            214               30.31
20      UASIN GISHU            1373            414               30.15
21            KITUI            1609            485               30.14
22             MERU            1034            310               29.98
23         KAKAMEGA            2274            681               29.95
24          NAIROBI            2866            858               29.94
25         MURANG'A             939            281               29.93
26  ELGEYO MARAKWET             718            212               29.53
27          KERICHO            1487            438               29.46
28          MAKUENI            1276            375               29.39
29           NAKURU            1801            524               29.09
30           MIGORI            2009            584               29.07
31       WEST POKOT             444            129               29.05
32        NYANDARUA             435            125               28.74
33            NANDI            1431            409               28.58
34         LAIKIPIA             355            101               28.45
35            KWALE             268             76               28.36
36            NYERI             658            186               28.27
37           ISIOLO              78             22               28.21
38          GARISSA              32              9               28.12
39           VIHIGA             706            195               27.62
40          KAJIADO             503            138               27.44
41          BARINGO             817            224               27.42
42          TURKANA              88             24               27.27
43          MOMBASA             561            152               27.09
44    THARAKA NITHI             373            100               26.81
```

```
45     TAITA TAVETA              204              46                22.55
46       TANA RIVER               66              13                19.70
```

[20]:
```python
# Sort by percentage highest to lowest
dropout_plot = dropout_table.sort_values('Dropout_Percentage', ascending=False).
 ↪reset_index(drop=True)

# Create colour gradient - green (low) to red (high)
norm = plt.Normalize(
    dropout_plot['Dropout_Percentage'].min(),
    dropout_plot['Dropout_Percentage'].max()
)
colors = cm.RdYlGn_r(norm(dropout_plot['Dropout_Percentage']))

# Create the figure
fig, ax = plt.subplots(figsize=(16, 8))

# Plot vertical bar chart with gradient colours
bars = ax.bar(
    dropout_plot['County'],
    dropout_plot['Dropout_Percentage'],
    color=colors,
    edgecolor='white',
    linewidth=0.5
)

# Add percentage labels on top of each bar
for bar, val in zip(bars, dropout_plot['Dropout_Percentage']):
    ax.text(
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.2,
        f'{val:.1f}%',
        ha='center',
        fontsize=7
    )

# Add colourbar legend
sm = cm.ScalarMappable(cmap='RdYlGn_r', norm=norm)
sm.set_array([])
cbar = plt.colorbar(sm, ax=ax, orientation='vertical', pad=0.02)
cbar.set_label('Dropout Percentage (%)', fontsize=10)

# Formatting
ax.set_ylabel('Dropout Percentage (%)', fontsize=12)
ax.set_xlabel('County', fontsize=12)
ax.set_title('School Dropout Rate by County in Kenya', fontsize=14,␣
 ↪fontweight='bold', pad=15)
```

```
ax.set_ylim(0, dropout_plot['Dropout_Percentage'].max() + 5)
ax.axhline(
    dropout_table['Dropout_Percentage'].mean(),
    color='navy', linestyle='--', linewidth=1.5,
    label=f"National Average: {dropout_table['Dropout_Percentage'].mean():.1f}%"
)
ax.legend(fontsize=10)
ax.grid(axis='y', linestyle='--', alpha=0.4)
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.xticks(rotation=90, fontsize=8)
plt.tight_layout()
plt.show()
```



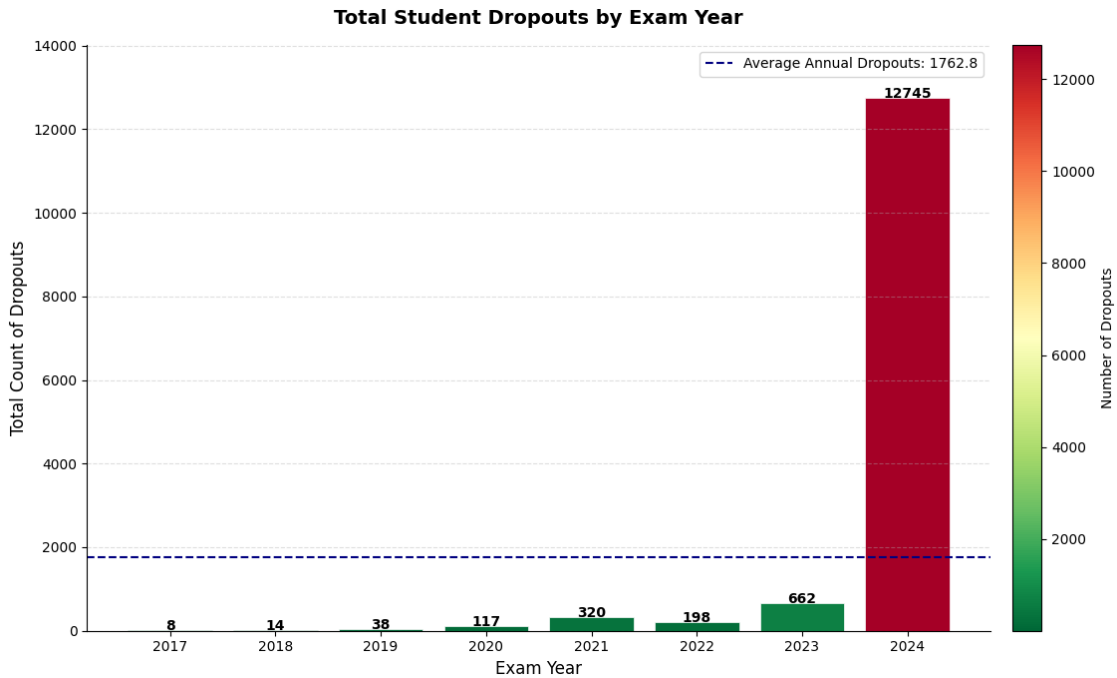
[21]:
```
#Create the summary table by ExamYear
dropout_year_table = School_df.groupby('ExamYear').agg(
    Total_Students = ('Drop_out', 'count'),
    Dropout_Count  = ('Drop_out', 'sum')
).reset_index()

# Calculate dropout percentage
dropout_year_table['Dropout_Percentage'] = (
    dropout_year_table['Dropout_Count'] / dropout_year_table['Total_Students']␣
 ↪* 100
).round(2)
```

```
# Sort by ExamYear
dropout_year_table = dropout_year_table.sort_values('ExamYear', ascending=True).
 ↪reset_index(drop=True)

print(dropout_year_table)
```

```
   ExamYear  Total_Students  Dropout_Count  Dropout_Percentage
0      2016               1              0                0.00
1      2017              18              8               44.44
2      2018              50             14               28.00
3      2019             146             38               26.03
4      2020             362            117               32.32
5      2021            1098            320               29.14
6      2022             680            198               29.12
7      2023            2273            662               29.12
8      2024           42255          12745               30.16
```

[22]:
```
import matplotlib.pyplot as plt
import matplotlib.cm as cm

# 1. Prepare the data: Count the number of dropouts per year
# We filter the dataframe for only dropouts and then group by year
dropout_counts = School_df[School_df['Drop_out'] == 1].groupby('ExamYear').
 ↪size().reset_index(name='Dropout_Count')

# 2. Plotting logic
fig, ax = plt.subplots(figsize=(12, 7))

# Normalization for the color map based on counts
norm = plt.Normalize(
    dropout_counts['Dropout_Count'].min(),
    dropout_counts['Dropout_Count'].max()
)
colors = cm.RdYlGn_r(norm(dropout_counts['Dropout_Count']))

bars = ax.bar(
    dropout_counts['ExamYear'].astype(str),
    dropout_counts['Dropout_Count'],
    color=colors,
    edgecolor='white',
    linewidth=0.5
)

# Add count labels on top of each bar
for bar, val in zip(bars, dropout_counts['Dropout_Count']):
    ax.text(
```

```
        bar.get_x() + bar.get_width() / 2,
        bar.get_height() + 0.5,
        f'{int(val)}',  # Displaying as integer count
        ha='center',
        fontsize=10,
        fontweight='bold'
    )

# Colourbar updated for counts
sm = cm.ScalarMappable(cmap='RdYlGn_r', norm=norm)
sm.set_array([])
cbar = plt.colorbar(sm, ax=ax, orientation='vertical', pad=0.02)
cbar.set_label('Number of Dropouts', fontsize=10)

# Labels and Title
ax.set_ylabel('Total Count of Dropouts', fontsize=12)
ax.set_xlabel('Exam Year', fontsize=12)
ax.set_title('Total Student Dropouts by Exam Year', fontsize=14,␣
 ↪fontweight='bold', pad=15)

# Update Y-limit to provide space for labels
ax.set_ylim(0, dropout_counts['Dropout_Count'].max() * 1.1)

# Horizontal line for the average annual count
avg_count = dropout_counts['Dropout_Count'].mean()
ax.axhline(
    avg_count,
    color='navy', linestyle='--', linewidth=1.5,
    label=f"Average Annual Dropouts: {avg_count:.1f}"
)

ax.legend(fontsize=10)
ax.grid(axis='y', linestyle='--', alpha=0.4)
ax.spines['top'].set_visible(False)
ax.spines['right'].set_visible(False)

plt.tight_layout()
plt.show()
```



[23]:
```
School_df.shape
```

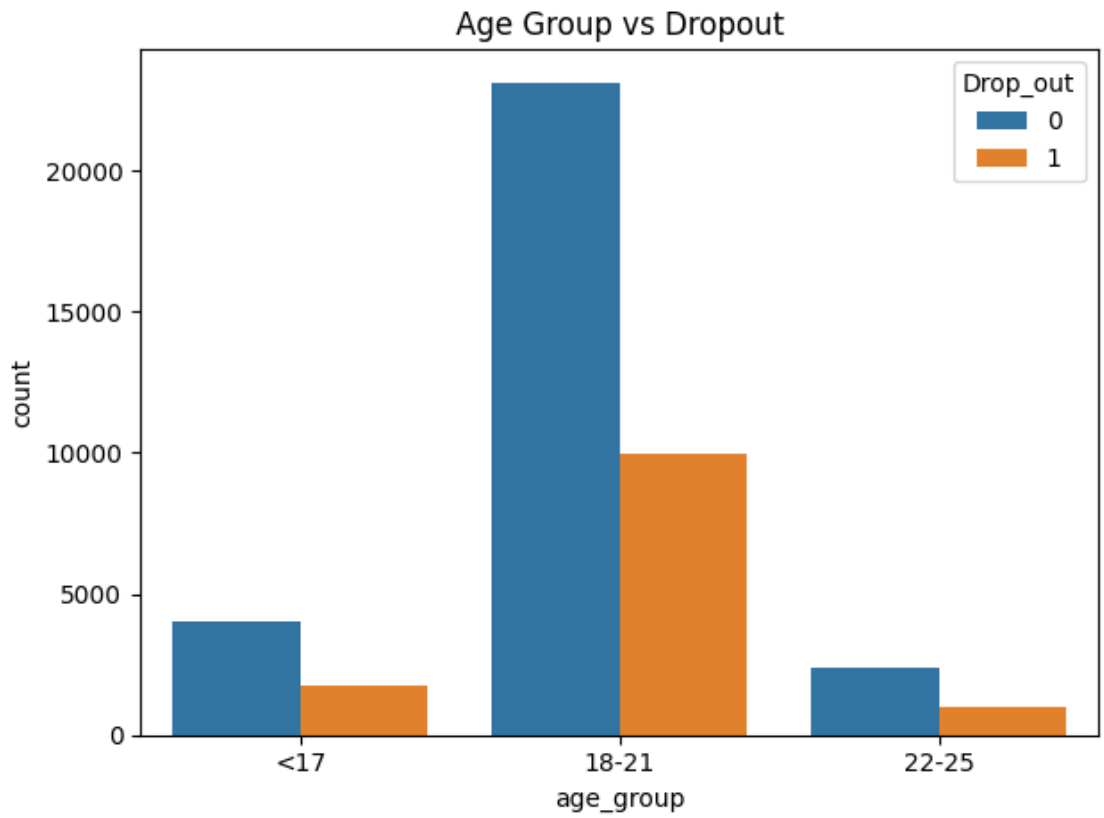
[23]: (46883, 22)

[24]:
```
# 1. (Optional but recommended) Ensure ExamYear is numeric to avoid type errors
School_df['ExamYear'] = pd.to_numeric(School_df['ExamYear'], errors='coerce')

# 2. Filter for the year 2024
School_df = School_df[School_df['ExamYear'] == 2024]

# 3. Check the results
print(f"Rows after filtering: {len(School_df)}")
print(f"Unique years in data: {School_df['ExamYear'].unique()}")
```

```
Rows after filtering: 42255
Unique years in data: [2024]
```

1.8.3 5.3 Age Group vs Dropout

[25]:
```
sns.countplot(x='age_group', hue='Drop_out', data=School_df)
plt.title('Age Group vs Dropout')
plt.tight_layout()
plt.show()
```

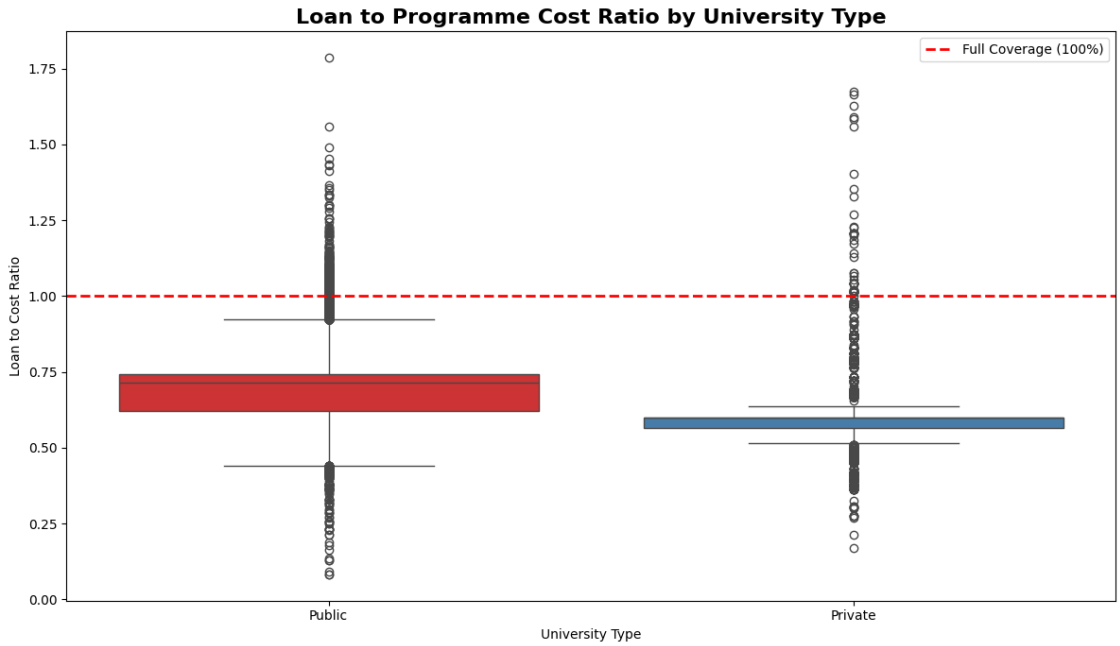


1.8.4 5.4 Loan-to-Cost Ratio by University Type

2 Remove Outliers

[26]:
```
df_ratio = School_df.dropna(subset=['ProgramCost', 'TotalLoanAllocated']).copy()
df_ratio = df_ratio[df_ratio['ProgramCost'] > 0]
df_ratio['LoanToCostRatio'] = (df_ratio['TotalLoanAllocated'] /␣
↪df_ratio['ProgramCost']).clip(upper=2)

plt.figure(figsize=(12, 7))
sns.boxplot(x='UniversityType', y='LoanToCostRatio', data=df_ratio,␣
↪palette='Set1')
plt.axhline(y=1, color='red', linestyle='--', linewidth=2, label='Full Coverage␣
↪(100%)')
plt.title('Loan to Programme Cost Ratio by University Type', fontsize=16,␣
↪fontweight='bold')
plt.xlabel('University Type')
plt.ylabel('Loan to Cost Ratio')
plt.legend()
plt.tight_layout()
```

```
plt.show()
```



[27]:
```
School_df.columns
```

[27]:
```
Index(['Gender', 'Birthdate', 'County', 'ScholarshipApplied',
       'InstitutionName', 'ProgramCost', 'CourseCategory', 'ExamYear',
       'TotalLoanAllocated', 'LoanproductCode', 'Category', 'LoanStatus',
       'Mother_Age', 'Father_Age', 'Father_educ_level', 'Mother_educ_level',
       'Highest_educ_level', 'Drop_out', 'age', 'UniversityType', 'Sponsored',
       'age_group'],
      dtype='str')
```

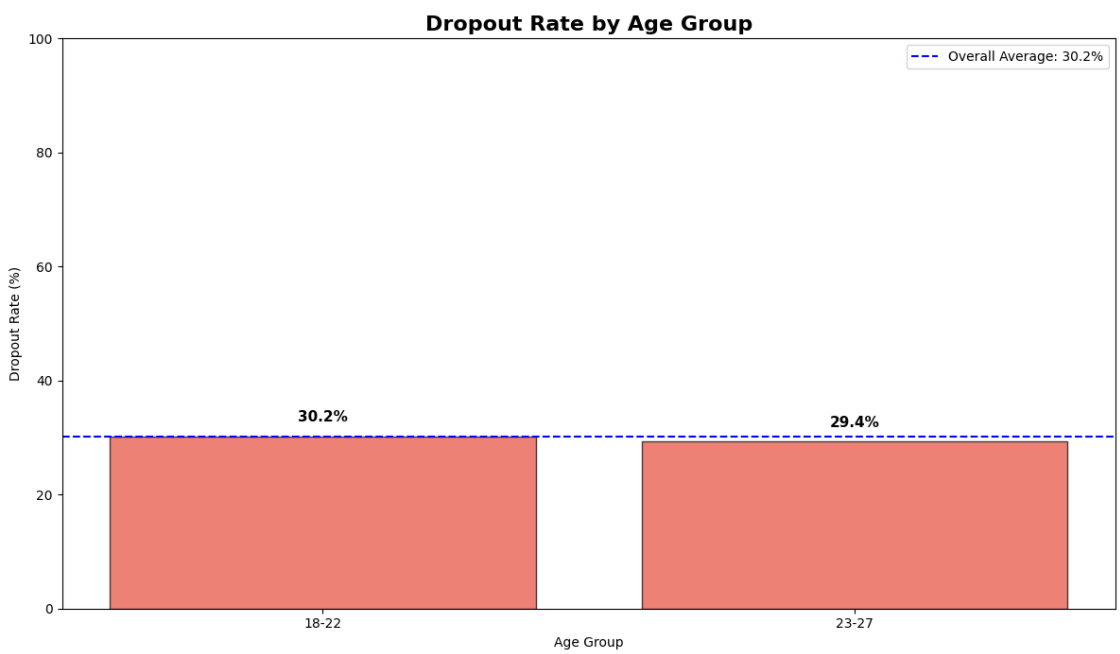
Interpretation: Public universities show a higher loan-to-cost ratio due to government support and lower fees. Private universities have a lower ratio, reflecting higher programme costs relative to loan allocations.

3 EDIT ME

3.0.1 5.6 Dropout Rate by Age Group

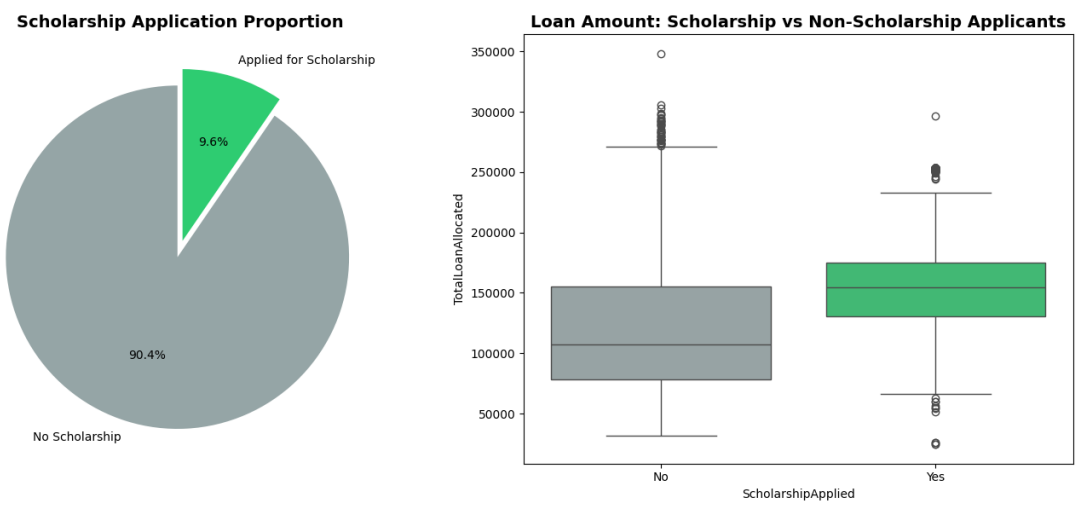
[28]:
```
df_age_group = School_df.dropna(subset=['age']).copy()
df_age_group = df_age_group[(df_age_group['age'] >= 18) & (df_age_group['age']␣
↪<= 60)]
bins2   = [18, 22, 27, 32, 40, 100]
labels2 = ['18-22', '23-27', '28-32', '33-40', '40+']
```

```
df_age_group['AgeGroup'] = pd.cut(df_age_group['age'], bins=bins2,␣
↪labels=labels2, right=False)
dropout_by_age = df_age_group.groupby('AgeGroup')['Drop_out'].agg(['mean',␣
↪'count'])
dropout_by_age = dropout_by_age[dropout_by_age['count'] > 10]

plt.figure(figsize=(12, 7))
bars = plt.bar(range(len(dropout_by_age)), dropout_by_age['mean'] * 100,
               color='#e74c3c', edgecolor='black', alpha=0.7)
plt.xticks(range(len(dropout_by_age)), dropout_by_age.index)
plt.title('Dropout Rate by Age Group', fontsize=16, fontweight='bold')
plt.xlabel('Age Group')
plt.ylabel('Dropout Rate (%)')
plt.ylim(0, 100)
for bar, rate in zip(bars, dropout_by_age['mean']):
    plt.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 2,
             f'{rate*100:.1f}%', ha='center', va='bottom', fontsize=11,␣
↪fontweight='bold')
overall = df_age_group['Drop_out'].mean() * 100
plt.axhline(y=overall, color='blue', linestyle='--', label=f'Overall Average:␣
↪{overall:.1f}%')
plt.legend()
plt.tight_layout()
plt.show()
```



3.0.2 5.7 Scholarship Application & Loan Amount

[29]:
```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))

scholarship_counts = School_df['ScholarshipApplied'].value_counts()
axes[0].pie(scholarship_counts.values,
            labels=['No Scholarship', 'Applied for Scholarship'],
            autopct='%1.1f%%', colors=['#95a5a6', '#2ecc71'],
            startangle=90, explode=(0.05, 0.05))
axes[0].set_title('Scholarship Application Proportion', fontsize=14,␣
 ↪fontweight='bold')

sns.boxplot(x='ScholarshipApplied', y='TotalLoanAllocated', data=School_df,
            ax=axes[1], palette=['#95a5a6', '#2ecc71'])
axes[1].set_title('Loan Amount: Scholarship vs Non-Scholarship Applicants',␣
 ↪fontsize=14, fontweight='bold')
axes[1].set_xticklabels(['No', 'Yes'])
plt.tight_layout()
plt.show()
```



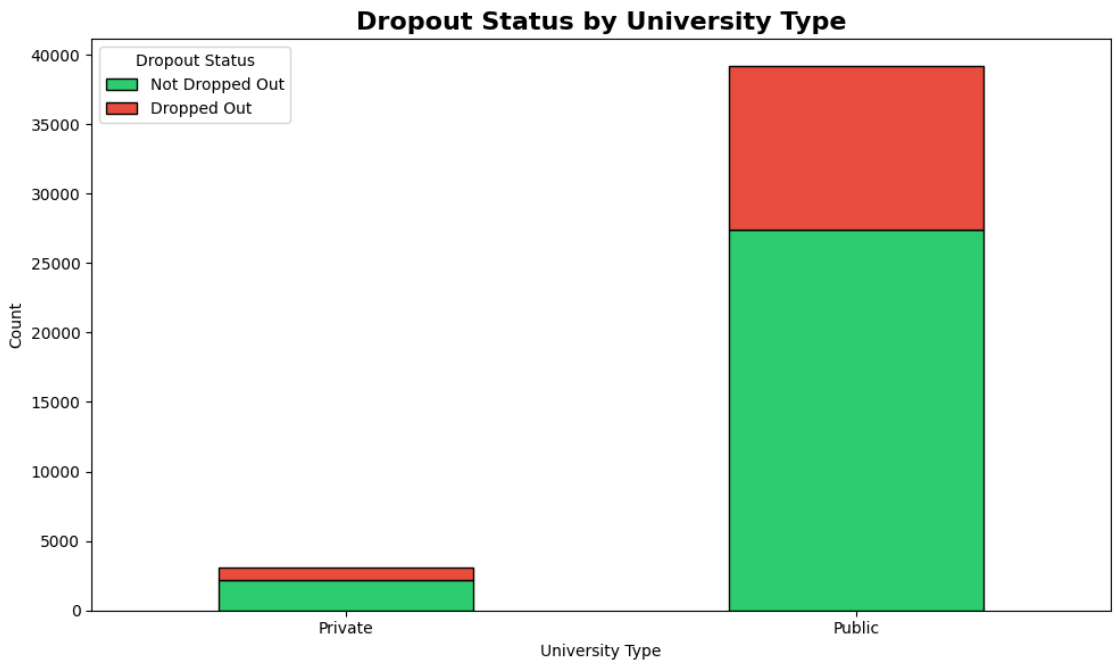
3.0.3 5.8 Loan Distribution by Demographic Groups

[30]:
```
fig, axes = plt.subplots(1, 3, figsize=(18, 6))

for ax, col, title in zip(axes,
                          ['LoanStatus', 'Gender', 'Sponsored'],
                          ['Loan Status Distribution', 'Loan Amount by␣
 ↪Gender', 'Loan Amount by Sponsorship']):
    if col == 'LoanStatus':
        counts = School_df[col].value_counts()
```

```
        ax.bar(counts.index, counts.values, edgecolor='black')
        ax.set_title(title, fontweight='bold')
        ax.tick_params(axis='x', rotation=45)
    else:
        sns.boxplot(x=col, y='TotalLoanAllocated', data=School_df, ax=ax,␣
 ↪palette='Set2')
        ax.set_title(title, fontweight='bold')
        ax.set_ylabel('Total Loan Allocated (KES)')

plt.tight_layout()
plt.show()
```

3.0.4 5.11 Dropout Status by University Type

[31]:
```
dropout_by_type = pd.crosstab(School_df['UniversityType'],␣
 ↪School_df['Drop_out'])
dropout_by_type.columns = ['Not Dropped Out', 'Dropped Out']
dropout_by_type.plot(kind='bar', stacked=True, color=['#2ecc71', '#e74c3c'],␣
 ↪edgecolor='black', figsize=(10, 6))
plt.title('Dropout Status by University Type', fontsize=16, fontweight='bold')
plt.xlabel('University Type')
plt.ylabel('Count')
plt.xticks(rotation=0)
plt.legend(title='Dropout Status')
plt.tight_layout()
plt.show()
```



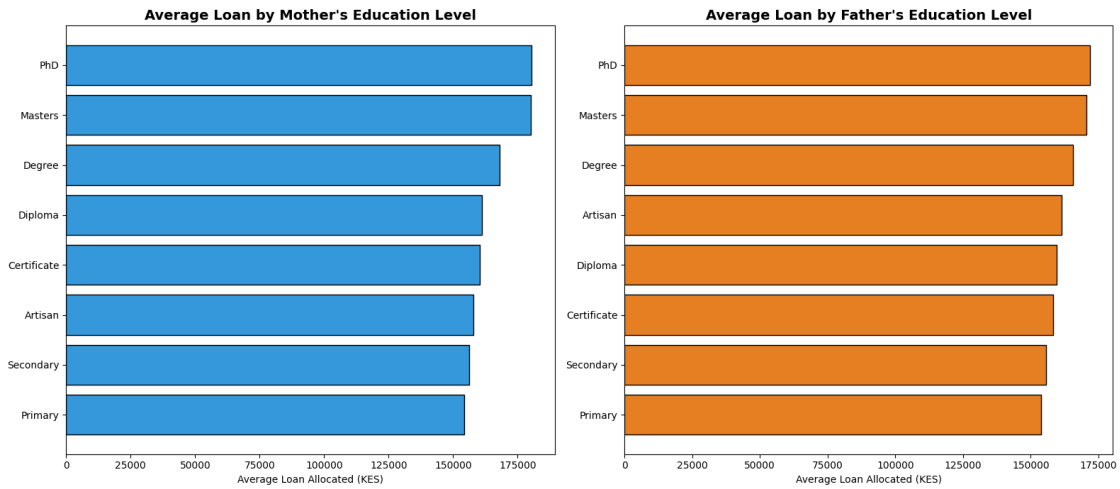
[32]:
```
# Filter the dataframe to exclude 'Private'
School_df = School_df[School_df['UniversityType'] != 'Private']

# Optional: Reset the index after dropping rows
School_df = School_df.reset_index(drop=True)

# Verification
print(f"Remaining rows: {len(School_df)}")
print(f"University types currently in data: {School_df['UniversityType'].
 ↪unique()}")
```

```
Remaining rows: 39166
University types currently in data: <StringArray>
['Public']
Length: 1, dtype: str
```

3.0.5 5.12 Average Loan by Parental Education Level

[33]:
```
fig, axes = plt.subplots(1, 2, figsize=(16, 7))
for ax, col, color, title in zip(
    axes,
    ['Mother_educ_level', 'Father_educ_level'],
    ['#3498db', '#e67e22'],
    ["Average Loan by Mother's Education Level", "Average Loan by Father's␣
 ↪Education Level"]
):
```

```
    edu = School_df.groupby(col)['TotalLoanAllocated'].mean().sort_values()
    ax.barh(edu.index, edu.values, color=color, edgecolor='black')
    ax.set_title(title, fontsize=14, fontweight='bold')
    ax.set_xlabel('Average Loan Allocated (KES)')
plt.tight_layout()
plt.show()
```



3.1 6. Hypothesis Tests

3.1.1 6.1 Scholarship Application vs Dropout

- **H :** No significant association between scholarship application and dropout status.
- **H :** Scholarship application is significantly associated with dropout status.

[34]:
```
table = pd.crosstab(School_df['ScholarshipApplied'], School_df['Drop_out'])
chi2, p, dof, expected = chi2_contingency(table)
print(f"Chi-square statistic: {chi2:.4f}")
print(f"p-value: {p:.4f}")
print("Decision:", "Reject H " if p < 0.05 else "Fail to reject H ")
```

```
Chi-square statistic: 0.0525
p-value: 0.8188
Decision: Fail to reject H
```

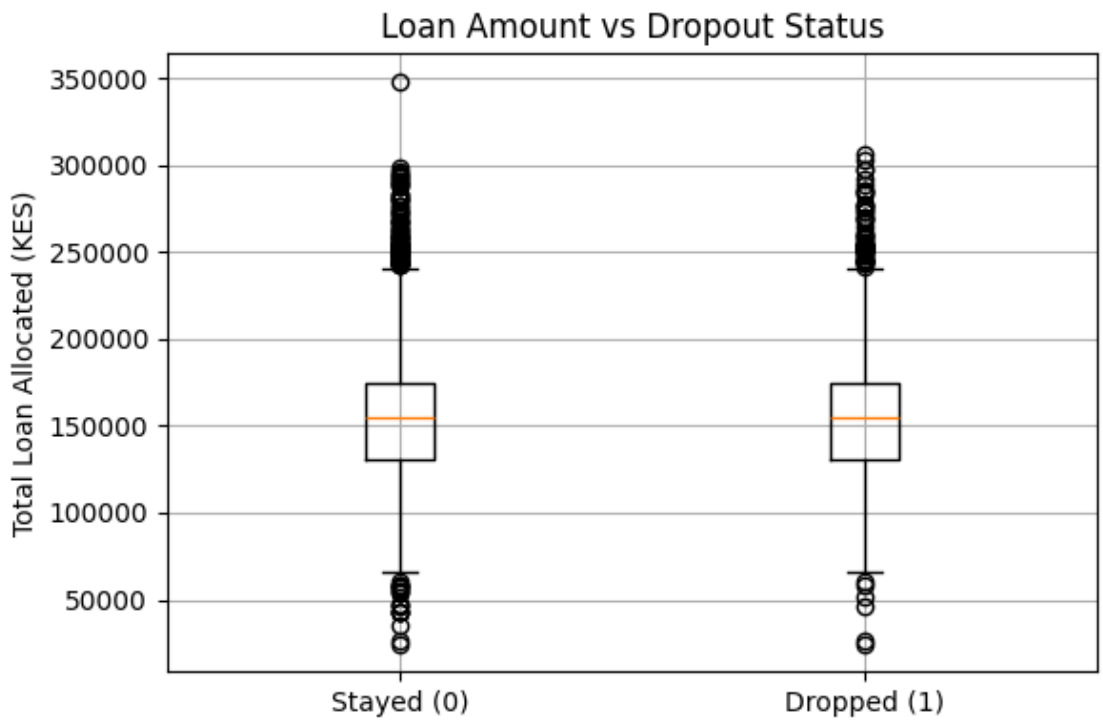
3.1.2 6.2 Loan Amount vs Dropout

- **H :** Loan amount has no effect on dropout status.
- **H :** Loan amount is significantly associated with dropout status.

[35]:
```
dropped = School_df[School_df['Drop_out'] == 1]['TotalLoanAllocated'].dropna()
stayed  = School_df[School_df['Drop_out'] == 0]['TotalLoanAllocated'].dropna()
t_stat, p_value = ttest_ind(dropped, stayed, equal_var=False)
```

```
print(f"T-statistic: {t_stat:.4f}")
print(f"P-value: {p_value:.4f}")
print("Decision:", "Reject H " if p_value < 0.05 else "Fail to reject H ")

plt.figure(figsize=(6, 4))
plt.boxplot([stayed, dropped], labels=['Stayed (0)', 'Dropped (1)'])
plt.title('Loan Amount vs Dropout Status')
plt.ylabel('Total Loan Allocated (KES)')
plt.grid(True)
plt.tight_layout()
plt.show()
```

```
T-statistic: -1.4771
P-value: 0.1397
Decision: Fail to reject H
```



3.1.3 6.3 Parental Education Levels vs Dropout

Testing father's, mother's, and highest parental education level against dropout status.

[36]:
```
for col, label in [
    ('Father_educ_level', "Father's Education"),
    ('Mother_educ_level', "Mother's Education"),
    ('Highest_educ_level', 'Highest Parental Education')
```

```
]:
    table = pd.crosstab(School_df[col], School_df['Drop_out'])
    chi2, p, dof, _ = chi2_contingency(table)
    decision = "Reject H " if p < 0.05 else "Fail to reject H "
    print(f"{label:<35} | chi2={chi2:.4f} | p={p:.4f} | {decision}")
```

```
Father's Education                  | chi2=8.3368 | p=0.3038 | Fail to reject H
Mother's Education                  | chi2=2.9436 | p=0.8902 | Fail to reject H
Highest Parental Education          | chi2=5.1070 | p=0.6469 | Fail to reject H
```

3.2 7. Modelling

3.2.1 7.1 Prepare Features and Target

[ ]:

```

```

[ ]:

```
# # Define features and target
# X = School_df.drop(columns=[
#     'Drop_out', 'Birthdate', 'InstitutionName',
#     'LoanproductCode', 'Mother_Age', 'Father_Age',
#     'Father_educ_level', 'Mother_educ_level', 'age_group', 'ExamYear',
#     'UniversityType'
# ], errors='ignore')

# y = School_df['Drop_out']

# print("Class distribution:")
# print(y.value_counts())
# print(f"\nDropout rate: {y.mean():.2%}")
```

```
Class distribution:
Drop_out
0    27359
1    11807
Name: count, dtype: int64

Dropout rate: 30.15%
```

[38]:

```
X.columns
```

[38]:

```
Index(['Gender', 'County', 'ScholarshipApplied', 'ProgramCost',
       'CourseCategory', 'TotalLoanAllocated', 'Category', 'LoanStatus',
       'Highest_educ_level', 'age', 'Sponsored'],
      dtype='str')
```

3.2.2 7.2 Encode and Split

Fix applied: `pd.get_dummies` is called on the full feature matrix **before** the train/test split. This ensures both sets share identical columns, eliminating the leaky `align(join='left')` pattern.

[39]:
```
# # Encode categoricals BEFORE splitting to ensure consistent columns
# X_encoded = pd.get_dummies(X)

# X_train, X_test, y_train, y_test = train_test_split(
#     X_encoded, y, train_size=0.8, random_state=452, stratify=y
# )

# print(f"Train: {X_train.shape}, Test: {X_test.shape}")
```

3.2.3 7.3 Define Models

[40]:
```
# LOGISTIC REGRESSION

# # Train
# lr = LogisticRegression(max_iter=1000, class_weight='balanced')
# lr.fit(X_train, y_train)

# # Predict
# y_pred = lr.predict(X_test)
# y_prob = lr.predict_proba(X_test)[:, 1]

# # Evaluate
# print("LOGISTIC REGRESSION")
# print("Accuracy:", accuracy_score(y_test, y_pred))
# print("ROC-AUC:", roc_auc_score(y_test, y_prob))
# print(classification_report(y_test, y_pred))
```

[ ]:
```
# # Check actual column names
# print(X_train.columns.tolist())
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[41], line 2
      1 # Check actual column names
----> 2 print(X_train.columns.tolist())

NameError: name 'X_train' is not defined
```

[ ]:
```
# # ##PIPELINE MODEL
# # Separate column types
# cat_cols = X.select_dtypes(include='object').columns
# num_cols = X.select_dtypes(exclude='object').columns

# # Preprocessing
# preprocessor = ColumnTransformer([
#     ('num', StandardScaler(), num_cols),
```

```
#     ('cat', OneHotEncoder(handle_unknown='ignore'), cat_cols)
# ])

# # Pipeline
# pipe = Pipeline([
#     ('preprocessor', preprocessor),
#     ('model', LogisticRegression(max_iter=1000))
# ])

# # Train
# pipe.fit(X_train, y_train)

# # Predict
# y_pred_pipe = pipe.predict(X_test)
# y_prob_pipe = pipe.predict_proba(X_test)[:, 1]

# # Evaluate
# print("\nPIPELINE LOGISTIC REGRESSION")
# print("Accuracy:", accuracy_score(y_test, y_pred_pipe))
# print("ROC-AUC:", roc_auc_score(y_test, y_prob_pipe))

# print(classification_report(y_test, y_pred_pipe))
```

[45]:

```
# Core libraries
import pandas as pd
import numpy as np
from datetime import datetime

# Visualisation
import matplotlib.pyplot as plt
import seaborn as sns
import matplotlib.cm as cm

# Statistics
from scipy.stats import chi2_contingency, ttest_ind

# Preprocessing
from sklearn.preprocessing import StandardScaler, LabelEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline

# Models
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier,␣
↪GradientBoostingClassifier, StackingClassifier
from imblearn.ensemble import BalancedRandomForestClassifier
from xgboost import XGBClassifier
```

```python
# Model selection & tuning
from sklearn.model_selection import (
    train_test_split, RandomizedSearchCV, StratifiedKFold, cross_val_score
)

# Metrics
from sklearn.metrics import (
    accuracy_score, classification_report, confusion_matrix,
    ConfusionMatrixDisplay, roc_auc_score, f1_score, recall_score,
    roc_curve, auc, precision_recall_curve, precision_score
)

# Warnings
import warnings
warnings.filterwarnings('ignore')
```

[42]:
```python
# Create additional features that may help prediction
# 1. Loan-to-Cost ratio (financial burden indicator)
School_df['LoanToCostRatio'] = (School_df['TotalLoanAllocated'] /␣
↪School_df['ProgramCost']).clip(upper=2)

# 2. Age at enrollment (calculated from Birthdate)
School_df['Birthdate'] = pd.to_datetime(School_df['Birthdate'], errors='coerce')
current_year = datetime.now().year
School_df['EnrollmentAge'] = current_year - School_df['Birthdate'].dt.year
School_df['EnrollmentAge'] = School_df['EnrollmentAge'].clip(16, 60)

# 3. Parental education composite (simple ordinal encoding)
edu_mapping = {'Primary': 1, 'Secondary': 2, 'Diploma': 3, 'Bachelor': 4,␣
↪'Masters': 5, 'PhD': 6}
for col in ['Father_educ_level', 'Mother_educ_level', 'Highest_educ_level']:
    School_df[f'{col}_encoded'] = School_df[col].map(edu_mapping).fillna(1)

# 4. Age group bins
bins = [16, 20, 25, 30, 40, 60]
labels = ['16-20', '21-25', '26-30', '31-40', '40+']
School_df['AgeGroup'] = pd.cut(School_df['EnrollmentAge'], bins=bins,␣
↪labels=labels)

print("Feature engineering complete")
print(f"Shape after feature engineering: {School_df.shape}")
```

```
Feature engineering complete
Shape after feature engineering: (39166, 28)
```

[70]:
```
# Define target
TARGET = 'Drop_out'

# Select features for modelling - use encoded versions for categoricals
FEATURE_COLS = [
    # Demographic
    'Gender', 'EnrollmentAge',
    # Financial
    'ProgramCost', 'TotalLoanAllocated', 'LoanToCostRatio',␣
 ↪'ScholarshipApplied',
    'LoanproductCode', 'LoanStatus',
    # Academic
    'ExamYear', 'CourseCategory',
    # Institutional
    'UniversityType', 'Sponsored',
    # Parental background (use encoded versions)
    'Father_educ_level_encoded', 'Mother_educ_level_encoded',␣
 ↪'Highest_educ_level_encoded',
    # Geographic
    'County'
]

# Keep only available columns
available_cols = [c for c in FEATURE_COLS if c in School_df.columns]
print(f"Available features: {len(available_cols)}")
print(available_cols)

# Prepare data
df_model = School_df[available_cols + [TARGET]].dropna().copy()
print(f"\nModelling dataset shape: {df_model.shape}")
print(f"Dropout rate: {df_model[TARGET].mean():.2%}")
```

```
Available features: 16
['Gender', 'EnrollmentAge', 'ProgramCost', 'TotalLoanAllocated',
'LoanToCostRatio', 'ScholarshipApplied', 'LoanproductCode', 'LoanStatus',
'ExamYear', 'CourseCategory', 'UniversityType', 'Sponsored',
'Father_educ_level_encoded', 'Mother_educ_level_encoded',
'Highest_educ_level_encoded', 'County']

Modelling dataset shape: (39166, 17)
Dropout rate: 30.15%
```

[71]:
```
# Identify categorical columns (exclude already encoded numeric ones)
cat_cols = df_model[available_cols].select_dtypes(include='object').columns.
 ↪tolist()
num_cols = df_model[available_cols].select_dtypes(include='number').columns.
 ↪tolist()
```

```
# Also exclude any columns that are already encoded
encoded_cols = ['Father_educ_level_encoded', 'Mother_educ_level_encoded',␣
↪'Highest_educ_level_encoded']
num_cols = [c for c in num_cols if c not in encoded_cols]

print(f"Numeric columns: {num_cols}")
print(f"Categorical columns: {cat_cols}")

# Encode categorical variables
df_encoded = df_model.copy()
le_dict = {}

for col in cat_cols:
    le = LabelEncoder()
    df_encoded[col] = le.fit_transform(df_encoded[col].astype(str))
    le_dict[col] = le

# Prepare features and target - all numeric now
X = df_encoded[available_cols]
y = df_encoded[TARGET]

print(f"\nFinal feature matrix shape: {X.shape}")
print(f"All features are now numeric: {X.dtypes.unique()}")
```

```
Numeric columns: ['EnrollmentAge', 'ProgramCost', 'TotalLoanAllocated',
'LoanToCostRatio', 'ScholarshipApplied', 'ExamYear']
Categorical columns: ['Gender', 'LoanproductCode', 'LoanStatus',
'CourseCategory', 'UniversityType', 'Sponsored', 'County']

Final feature matrix shape: (39166, 16)
All features are now numeric: [dtype('int64') dtype('int32') dtype('float64')]
```

[72]:
```
# Stratified split to maintain class distribution
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.20, random_state=42, stratify=y
)

# Scale numeric features - create separate scaled versions for models that need␣
↪scaling
scaler = StandardScaler()

# For Logistic Regression (needs scaled data)
X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```

```
# For tree-based models (Random Forest, XGBoost, etc.), keep original unscaled␣
↪data
# They don't need scaling

print(f"Training set: {X_train.shape[0]} samples")
print(f"Test set: {X_test.shape[0]} samples")
print(f"Training dropout rate: {y_train.mean():.2%}")
print(f"Test dropout rate: {y_test.mean():.2%}")
print(f"\nScaled data shape: {X_train_scaled.shape}")
```

```
Training set: 31332 samples
Test set: 7834 samples
Training dropout rate: 30.14%
Test dropout rate: 30.15%

Scaled data shape: (31332, 16)
```

[73]:
```
def evaluate_model(name, model, X_tr, y_tr, X_te, y_te, results_dict,␣
↪use_scaled=False):
    """Train and evaluate a model, store results"""
    model.fit(X_tr, y_tr)
    y_pred = model.predict(X_te)
    y_proba = model.predict_proba(X_te)[:, 1]

    results_dict[name] = {
        'model': model,
        'y_pred': y_pred,
        'y_proba': y_proba,
        'Accuracy': accuracy_score(y_te, y_pred),
        'Precision': precision_score(y_te, y_pred),
        'Recall': recall_score(y_te, y_pred),
        'F1': f1_score(y_te, y_pred),
        'ROC-AUC': roc_auc_score(y_te, y_proba)
    }

    print(f"\n{'='*50}")
    print(f"  {name}")
    print(f"{'='*50}")
    print(f"  Accuracy:  {results_dict[name]['Accuracy']:.4f}")
    print(f"  Precision: {results_dict[name]['Precision']:.4f}")
    print(f"  Recall:    {results_dict[name]['Recall']:.4f}")
    print(f"  F1 Score:  {results_dict[name]['F1']:.4f}")
    print(f"  ROC-AUC:   {results_dict[name]['ROC-AUC']:.4f}")
    print("\nClassification Report:")
    print(classification_report(y_te, y_pred, target_names=['Stayed', 'Dropped␣
↪Out']))
```

```
    return model

results = {}
```

[74]:
```
# Calculate class weight
neg, pos = (y_train == 0).sum(), (y_train == 1).sum()
class_weight = {0: 1, 1: neg/pos}

# Logistic Regression with tuned parameters
lr_model = LogisticRegression(
    C=0.5,
    class_weight=class_weight,
    solver='liblinear',
    max_iter=1000,
    random_state=42
)

evaluate_model('Logistic Regression', lr_model,
               X_train_scaled, y_train, X_test_scaled, y_test, results)
```

```
==================================================
  Logistic Regression
==================================================
  Accuracy:  0.5040
  Precision: 0.3080
  Recall:    0.5174
  F1 Score:  0.3861
  ROC-AUC:   0.5109

Classification Report:
              precision    recall  f1-score   support

      Stayed       0.71      0.50      0.58      5472
 Dropped Out       0.31      0.52      0.39      2362

    accuracy                           0.50      7834
   macro avg       0.51      0.51      0.48      7834
weighted avg       0.59      0.50      0.52      7834
```

[74]:
```
LogisticRegression(C=0.5, class_weight={0: 1, 1: np.float64(2.317310746426681)},
                   max_iter=1000, random_state=42, solver='liblinear')
```

[76]:
```
# Lighter Random Forest - Reduced search space and iterations
print("\n" + "="*50)
print("  Hyperparameter Tuning: Random Forest (Light)")
print("="*50)
```

```python
# Reduced parameter space
param_dist_rf = {
    'n_estimators': [100, 200],  # Reduced from 4 options to 2
    'max_depth': [10, 20],       # Removed None and 5,15
    'min_samples_split': [2, 5], # Reduced from 3 to 2
    'min_samples_leaf': [1, 2],  # Reduced from 3 to 2
    'max_features': ['sqrt', 0.5], # Reduced from 4 to 2
    'class_weight': ['balanced']   # Only one option
}

rf_base = RandomForestClassifier(random_state=42, n_jobs=-1)

# Reduce iterations from 30 to 10, cv from 5 to 3
rf_search = RandomizedSearchCV(
    rf_base,
    param_distributions=param_dist_rf,
    n_iter=10,           # Was 30
    cv=3,                # Was 5
    scoring='roc_auc',
    random_state=42,
    n_jobs=-1,
    verbose=0            # Turn off verbose output
)

# Use ORIGINAL (unscaled) data for tree-based models
rf_search.fit(X_train, y_train)

print(f"Best params: {rf_search.best_params_}")
print(f"Best CV ROC-AUC: {rf_search.best_score_:.4f}")

best_rf = rf_search.best_estimator_
evaluate_model('Random Forest (Tuned)', best_rf,
               X_train, y_train, X_test, y_test, results)
```

```
==================================================
  Hyperparameter Tuning: Random Forest (Light)
==================================================
Best params: {'n_estimators': 100, 'min_samples_split': 2, 'min_samples_leaf':
1, 'max_features': 0.5, 'max_depth': 10, 'class_weight': 'balanced'}
Best CV ROC-AUC: 0.4970

==================================================
  Random Forest (Tuned)
==================================================
  Accuracy:  0.5479
  Precision: 0.3110
```

```
  Recall:    0.4111
  F1 Score:  0.3541
  ROC-AUC:   0.5141

Classification Report:
              precision    recall  f1-score   support

      Stayed       0.70      0.61      0.65      5472
 Dropped Out       0.31      0.41      0.35      2362

    accuracy                           0.55      7834
   macro avg       0.51      0.51      0.50      7834
weighted avg       0.59      0.55      0.56      7834
```

[76]:
```
RandomForestClassifier(class_weight='balanced', max_depth=10, max_features=0.5,
                       n_jobs=-1, random_state=42)
```

[77]:
```python
print("\n" + "="*50)
print("  Hyperparameter Tuning: XGBoost")
print("="*50)

# Calculate scale_pos_weight for imbalance
scale_pos_weight = neg / pos

param_dist_xgb = {
    'n_estimators': [100, 200],
    'max_depth': [3, 5, 7],                # Was 6 options
    'learning_rate': [0.05, 0.1],          # Was 4 options
    'subsample': [0.8, 1.0],               # Was 5 options
    'colsample_bytree': [0.8, 1.0],        # Was 5 options
    'min_child_weight': [1, 3],            # Was 4 options
    'gamma': [0, 0.1],                     # Was 5 options
    'reg_alpha': [0, 0.1],                 # Was 4 options
    'reg_lambda': [1, 2]
}

xgb_base = XGBClassifier(
    scale_pos_weight=scale_pos_weight,
    eval_metric='logloss',
    random_state=42,
    n_jobs=-1
)

xgb_search = RandomizedSearchCV(
    xgb_base,
    param_distributions=param_dist_xgb,
```

```python
    n_iter=50,
    cv=5,
    scoring='roc_auc',
    random_state=42,
    n_jobs=-1,
    verbose=1
)

xgb_search.fit(X_train, y_train)

print(f"\nBest parameters: {xgb_search.best_params_}")
print(f"Best CV ROC-AUC: {xgb_search.best_score_:.4f}")

best_xgb = xgb_search.best_estimator_
evaluate_model('XGBoost (Tuned)', best_xgb,
               X_train, y_train, X_test, y_test, results)
```

```
==================================================
  Hyperparameter Tuning: XGBoost
==================================================
Fitting 5 folds for each of 50 candidates, totalling 250 fits

Best parameters: {'subsample': 1.0, 'reg_lambda': 2, 'reg_alpha': 0,
'n_estimators': 200, 'min_child_weight': 1, 'max_depth': 3, 'learning_rate':
0.1, 'gamma': 0.1, 'colsample_bytree': 0.8}
Best CV ROC-AUC: 0.5001

==================================================
  XGBoost (Tuned)
==================================================
  Accuracy:  0.4997
  Precision: 0.3099
  Recall:    0.5373
  F1 Score:  0.3931
  ROC-AUC:   0.5128

Classification Report:
              precision    recall  f1-score   support

      Stayed       0.71      0.48      0.57      5472
 Dropped Out       0.31      0.54      0.39      2362

    accuracy                           0.50      7834
   macro avg       0.51      0.51      0.48      7834
weighted avg       0.59      0.50      0.52      7834
```

[77]:
```
XGBClassifier(base_score=None, booster=None, callbacks=None,
              colsample_bylevel=None, colsample_bynode=None,
              colsample_bytree=0.8, device=None, early_stopping_rounds=None,
              enable_categorical=False, eval_metric='logloss',
              feature_types=None, feature_weights=None, gamma=0.1,
              grow_policy=None, importance_type=None,
              interaction_constraints=None, learning_rate=0.1, max_bin=None,
              max_cat_threshold=None, max_cat_to_onehot=None,
              max_delta_step=None, max_depth=3, max_leaves=None,
              min_child_weight=1, missing=nan, monotone_constraints=None,
              multi_strategy=None, n_estimators=200, n_jobs=-1,
              num_parallel_tree=None, …)
```

[78]:
```
# Balanced Random Forest - specifically designed for imbalanced data
brf_model = BalancedRandomForestClassifier(
    n_estimators=200,
    max_depth=10,
    min_samples_leaf=2,
    random_state=42,
    n_jobs=-1
)

evaluate_model('Balanced Random Forest', brf_model,
               X_train, y_train, X_test, y_test, results)
```

```
==================================================
  Balanced Random Forest
==================================================
  Accuracy:  0.5244
  Precision: 0.3094
  Recall:    0.4687
  F1 Score:  0.3727
  ROC-AUC:   0.5150

Classification Report:
              precision    recall  f1-score   support

      Stayed       0.71      0.55      0.62      5472
 Dropped Out       0.31      0.47      0.37      2362

    accuracy                           0.52      7834
   macro avg       0.51      0.51      0.49      7834
weighted avg       0.59      0.52      0.54      7834
```

[78]:
```
BalancedRandomForestClassifier(max_depth=10, min_samples_leaf=2,
                               n_estimators=200, n_jobs=-1, random_state=42)
```

[79]:
```
# Gradient Boosting Classifier
gbm_model = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.05,
    max_depth=4,
    min_samples_leaf=5,
    subsample=0.8,
    random_state=42
)

evaluate_model('Gradient Boosting', gbm_model,
               X_train, y_train, X_test, y_test, results)
```

```
==================================================
  Gradient Boosting
==================================================
  Accuracy:  0.6977
  Precision: 0.2857
  Recall:    0.0017
  F1 Score:  0.0034
  ROC-AUC:   0.5077

Classification Report:
              precision    recall  f1-score   support

      Stayed       0.70      1.00      0.82      5472
 Dropped Out       0.29      0.00      0.00      2362

    accuracy                           0.70      7834
   macro avg       0.49      0.50      0.41      7834
weighted avg       0.57      0.70      0.58      7834
```

[79]:
```
GradientBoostingClassifier(learning_rate=0.05, max_depth=4, min_samples_leaf=5,
                           n_estimators=200, random_state=42, subsample=0.8)
```

[80]:
```
print("\n" + "="*50)
print("  Building Stacking Ensemble")
print("="*50)

# Base learners (should be diverse)
base_learners = [
    ('lr', LogisticRegression(C=0.5, class_weight=class_weight,␣
 ↪solver='liblinear', max_iter=1000)),
    ('rf', best_rf),
    ('xgb', best_xgb),
```

```
    ('brf', BalancedRandomForestClassifier(n_estimators=200, max_depth=10,␣
 ↪random_state=42)),
    ('gbm', GradientBoostingClassifier(n_estimators=150, learning_rate=0.05,␣
 ↪max_depth=4, random_state=42))
]

# Meta-learner
meta_learner = LogisticRegression(
    C=1.0,
    class_weight='balanced',
    solver='liblinear',
    max_iter=1000
)

stacking_clf = StackingClassifier(
    estimators=base_learners,
    final_estimator=meta_learner,
    cv=5,
    stack_method='predict_proba',
    n_jobs=-1
)

evaluate_model('Stacking Ensemble', stacking_clf,
               X_train_scaled, y_train, X_test_scaled, y_test, results)
```

```
==================================================
  Building Stacking Ensemble
==================================================

==================================================
  Stacking Ensemble
==================================================
  Accuracy:  0.5525
  Precision: 0.2982
  Recall:    0.3577
  F1 Score:  0.3253
  ROC-AUC:   0.4902

Classification Report:
              precision    recall  f1-score   support

      Stayed       0.70      0.64      0.67      5472
 Dropped Out       0.30      0.36      0.33      2362

    accuracy                           0.55      7834
   macro avg       0.50      0.50      0.50      7834
```

```
weighted avg       0.58      0.55      0.56      7834
```

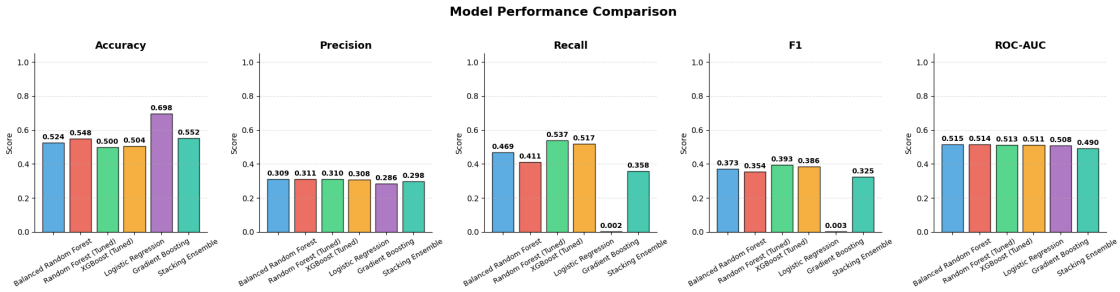
[80]:
```
StackingClassifier(cv=5,
                   estimators=[('lr',
                                LogisticRegression(C=0.5,
                                                   class_weight={0: 1,
                                                                 1:
np.float64(2.317310746426681)},
                                                   max_iter=1000,
                                                   solver='liblinear')),
                               ('rf',
                                RandomForestClassifier(class_weight='balanced',
                                                       max_depth=10,
                                                       max_features=0.5,
                                                       n_jobs=-1,
                                                       random_state=42)),
                               ('xgb',
                                XGBClassifier(base_score=None, booster=None,
                                              callbacks=None,
                                              colsample_bylevel=Non…
                                              n_estimators=200, n_jobs=-1,
                                              num_parallel_tree=None, …)),
                               ('brf',
                                BalancedRandomForestClassifier(max_depth=10,
                                                               n_estimators=200,
random_state=42)),
                               ('gbm',
                                GradientBoostingClassifier(learning_rate=0.05,
                                                           max_depth=4,
                                                           n_estimators=150,
                                                           random_state=42))],
                   final_estimator=LogisticRegression(class_weight='balanced',
                                                      max_iter=1000,
                                                      solver='liblinear'),
                   n_jobs=-1, stack_method='predict_proba')
```

[81]:
```python
# Create comparison dataframe
metrics = ['Accuracy', 'Precision', 'Recall', 'F1', 'ROC-AUC']
comparison_df = pd.DataFrame({
    name: {m: res[m] for m in metrics}
    for name, res in results.items()
}).T

# Sort by ROC-AUC (best first)
comparison_df = comparison_df.sort_values('ROC-AUC', ascending=False)
```

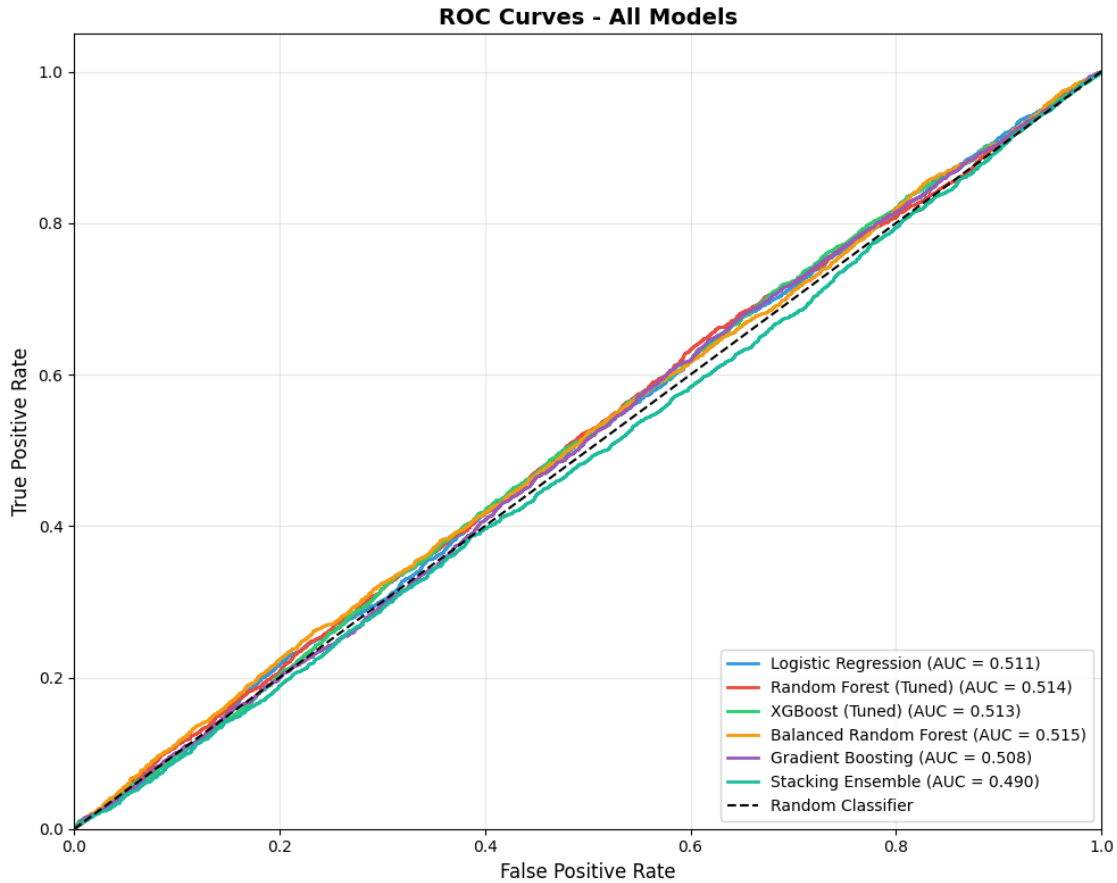
```
print("\n" + "="*70)
print("  MODEL COMPARISON SUMMARY")
print("="*70)
print(comparison_df.round(4).to_string())
print("="*70)
```

```
======================================================================
  MODEL COMPARISON SUMMARY
======================================================================
                        Accuracy  Precision  Recall      F1  ROC-AUC
Balanced Random Forest    0.5244     0.3094  0.4687  0.3727   0.5150
Random Forest (Tuned)     0.5479     0.3110  0.4111  0.3541   0.5141
XGBoost (Tuned)           0.4997     0.3099  0.5373  0.3931   0.5128
Logistic Regression       0.5040     0.3080  0.5174  0.3861   0.5109
Gradient Boosting         0.6977     0.2857  0.0017  0.0034   0.5077
Stacking Ensemble         0.5525     0.2982  0.3577  0.3253   0.4902
======================================================================
```

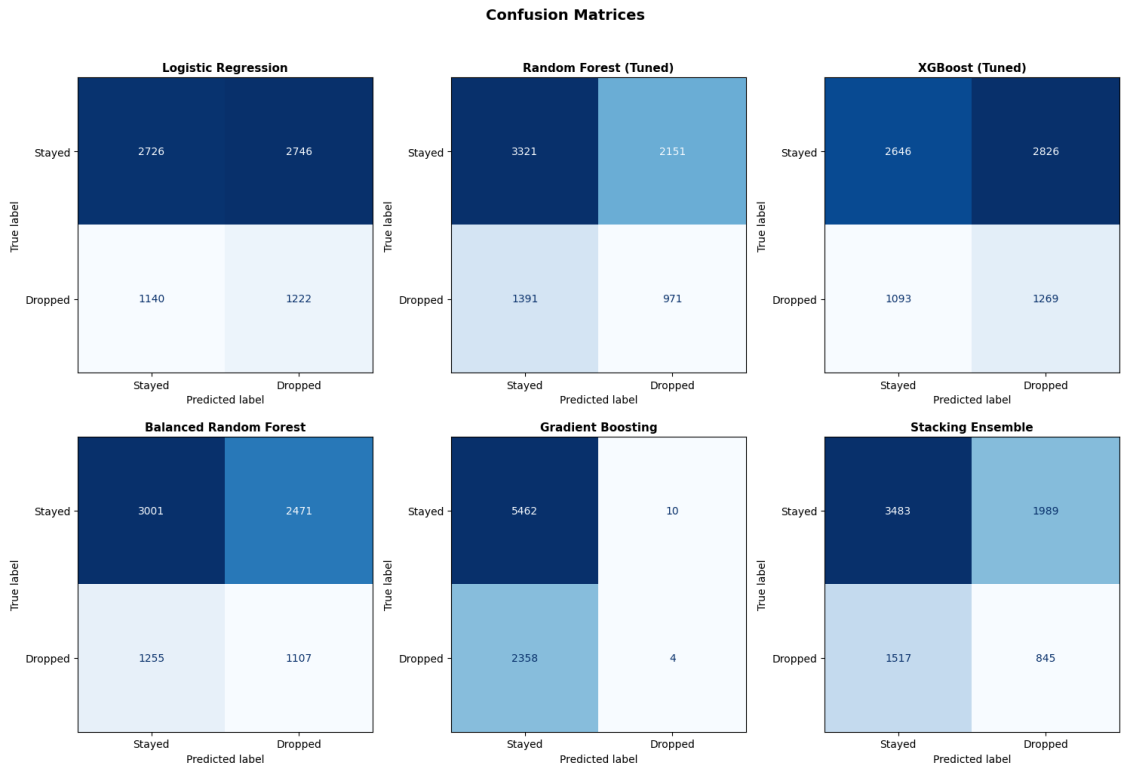
[82]:
```
fig, axes = plt.subplots(1, 5, figsize=(20, 5))
colors = ['#3498db', '#e74c3c', '#2ecc71', '#f39c12', '#9b59b6', '#1abc9c']

for ax, metric in zip(axes, metrics):
    values = comparison_df[metric].values
    labels = comparison_df.index.tolist()

    bars = ax.bar(labels, values, color=colors[:len(labels)],␣
 ↪edgecolor='black', alpha=0.8)
    ax.set_title(metric, fontsize=13, fontweight='bold')
    ax.set_ylim(0, 1.05)
    ax.set_ylabel('Score')
    ax.tick_params(axis='x', rotation=30, labelsize=9)
    ax.grid(axis='y', linestyle='--', alpha=0.4)
    ax.spines['top'].set_visible(False)
    ax.spines['right'].set_visible(False)

    # Add value labels on bars
    for bar, val in zip(bars, values):
        ax.text(bar.get_x() + bar.get_width()/2, bar.get_height() + 0.01,
                f'{val:.3f}', ha='center', va='bottom', fontsize=9,␣
 ↪fontweight='bold')

plt.suptitle('Model Performance Comparison', fontsize=16, fontweight='bold',␣
 ↪y=1.02)
plt.tight_layout()
plt.show()
```



[83]:
```
plt.figure(figsize=(10, 8))
colors_roc = ['#3498db', '#e74c3c', '#2ecc71', '#f39c12', '#9b59b6', '#1abc9c']

for (name, res), color in zip(results.items(), colors_roc):
    fpr, tpr, _ = roc_curve(y_test, res['y_proba'])
    roc_auc = res['ROC-AUC']
    plt.plot(fpr, tpr, lw=2, color=color, label=f'{name} (AUC = {roc_auc:.3f})')

plt.plot([0, 1], [0, 1], 'k--', lw=1.5, label='Random Classifier')
plt.xlim([0.0, 1.0])
plt.ylim([0.0, 1.05])
plt.xlabel('False Positive Rate', fontsize=12)
plt.ylabel('True Positive Rate', fontsize=12)
plt.title('ROC Curves - All Models', fontsize=14, fontweight='bold')
plt.legend(loc='lower right', fontsize=10)
plt.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```



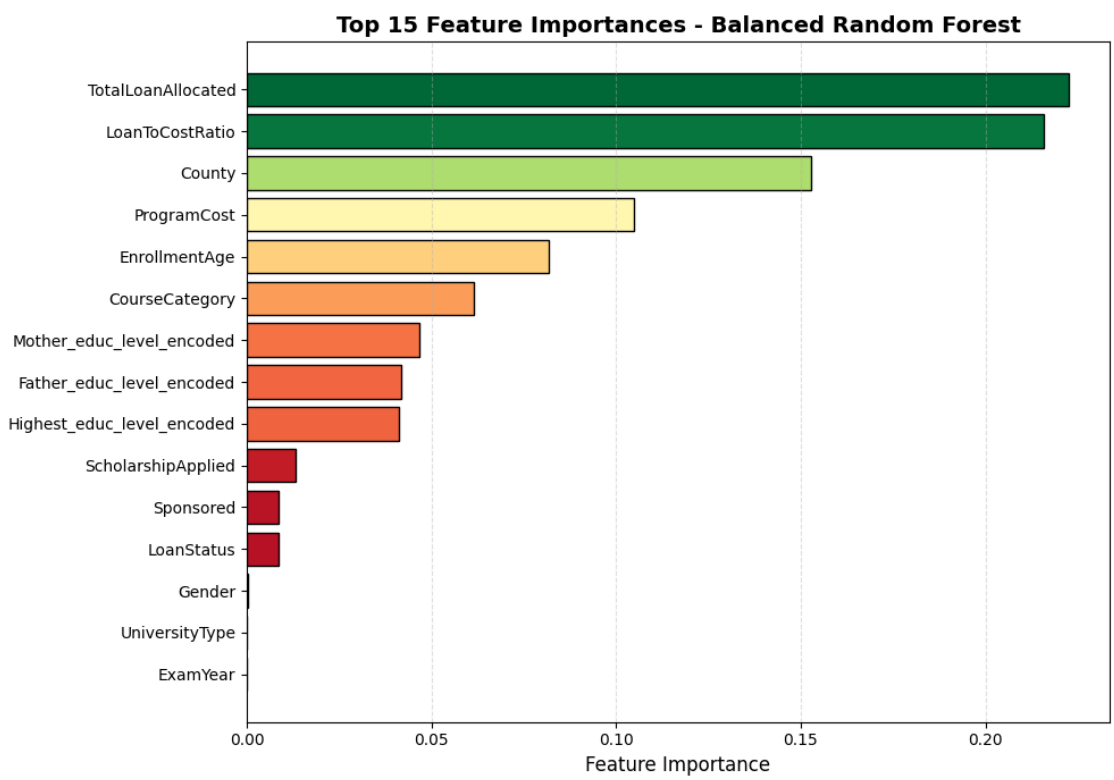
[84]:
```
n_models = len(results)
fig, axes = plt.subplots(2, 3, figsize=(15, 10))
axes = axes.flatten()

for idx, (name, res) in enumerate(results.items()):
    cm = confusion_matrix(y_test, res['y_pred'])
    disp = ConfusionMatrixDisplay(confusion_matrix=cm,␣
 ↪display_labels=['Stayed', 'Dropped'])
    disp.plot(ax=axes[idx], colorbar=False, cmap='Blues')
    axes[idx].set_title(name, fontsize=11, fontweight='bold')

# Hide unused subplot
for idx in range(len(results), len(axes)):
    axes[idx].set_visible(False)

plt.suptitle('Confusion Matrices', fontsize=14, fontweight='bold', y=1.02)
plt.tight_layout()
plt.show()
```



[85]:
```
# Get the best model based on ROC-AUC
best_model_name = comparison_df.index[0]
best_model = results[best_model_name]['model']

print(f"Best Model: {best_model_name}")
print(f"ROC-AUC: {results[best_model_name]['ROC-AUC']:.4f}")
print(f"Recall: {results[best_model_name]['Recall']:.4f}")

# Extract feature importance if available
if hasattr(best_model, 'feature_importances_'):
    importances = pd.Series(
        best_model.feature_importances_,
        index=X.columns
    ).sort_values(ascending=True).tail(15)

    plt.figure(figsize=(10, 7))
    colors_fi = plt.cm.RdYlGn(importances.values / importances.values.max())
    plt.barh(importances.index, importances.values, color=colors_fi,␣
↪edgecolor='black')
    plt.xlabel('Feature Importance', fontsize=12)
    plt.title(f'Top 15 Feature Importances - {best_model_name}', fontsize=14,␣
↪fontweight='bold')
```

```
    plt.grid(axis='x', linestyle='--', alpha=0.4)
    plt.tight_layout()
    plt.show()
elif hasattr(best_model, 'coef_'):
    # For logistic regression
    importances = pd.Series(
        np.abs(best_model.coef_[0]),
        index=X.columns
    ).sort_values(ascending=True).tail(15)

    plt.figure(figsize=(10, 7))
    plt.barh(importances.index, importances.values, color='#3498db',␣
 ↪edgecolor='black')
    plt.xlabel('Absolute Coefficient Magnitude', fontsize=12)
    plt.title(f'Top 15 Feature Coefficients - {best_model_name}', fontsize=14,␣
 ↪fontweight='bold')
    plt.grid(axis='x', linestyle='--', alpha=0.4)
    plt.tight_layout()
    plt.show()
```

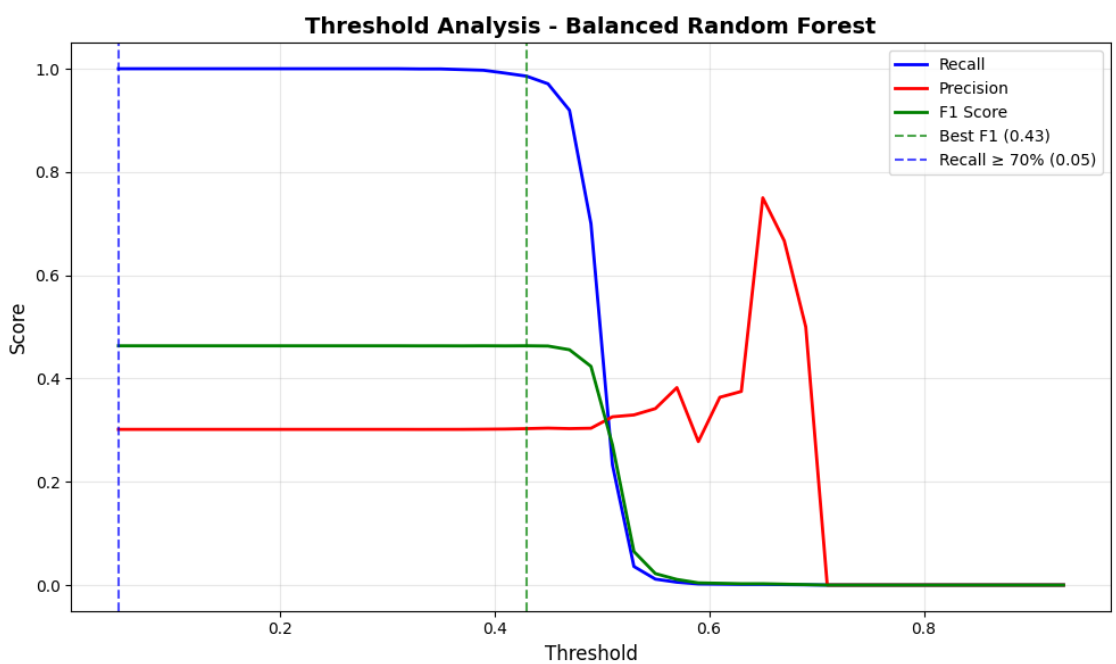
```
Best Model: Balanced Random Forest
ROC-AUC: 0.5150
Recall: 0.4687
```



[ ]:
```
# For the best model, find threshold that optimizes recall while maintaining␣
↪reasonable precision
best_model = results[best_model_name]['model']
y_proba = results[best_model_name]['y_proba']

thresholds = np.arange(0.05, 0.95, 0.02)
recall_scores = []
precision_scores = []
f1_scores = []

for t in thresholds:
    preds = (y_proba > t).astype(int)
    recall_scores.append(recall_score(y_test, preds))
    precision_scores.append(precision_score(y_test, preds))
    f1_scores.append(f1_score(y_test, preds))

# Find threshold that balances recall and precision
best_f1_idx = np.argmax(f1_scores)
best_f1_threshold = thresholds[best_f1_idx]

# find threshold for recall >= 0.70
recall_70_candidates = [(t, r, p) for t, r, p in zip(thresholds, recall_scores,␣
↪precision_scores) if r >= 0.70]
if recall_70_candidates:
    recall_70_threshold = recall_70_candidates[0][0]
    recall_70_recall = recall_70_candidates[0][1]
    recall_70_precision = recall_70_candidates[0][2]
else:
    recall_70_threshold = None

print(f"\n{'='*50}")
print("  THRESHOLD OPTIMIZATION")
print(f"{'='*50}")
print(f"Best F1 threshold: {best_f1_threshold:.2f}")
print(f"  Recall: {recall_scores[best_f1_idx]:.4f}, Precision:␣
↪{precision_scores[best_f1_idx]:.4f}")

if recall_70_threshold:
    print(f"\nThreshold for Recall  70%: {recall_70_threshold:.2f}")
    print(f"  Recall: {recall_70_recall:.4f}, Precision: {recall_70_precision:.
↪4f}")

# Plot precision-recall vs threshold
fig, ax = plt.subplots(figsize=(10, 6))
ax.plot(thresholds, recall_scores, 'b-', linewidth=2, label='Recall')
```

```
ax.plot(thresholds, precision_scores, 'r-', linewidth=2, label='Precision')
ax.plot(thresholds, f1_scores, 'g-', linewidth=2, label='F1 Score')
ax.axvline(x=best_f1_threshold, color='g', linestyle='--', alpha=0.7,␣
 ↪label=f'Best F1 ({best_f1_threshold:.2f})')
if recall_70_threshold:
    ax.axvline(x=recall_70_threshold, color='b', linestyle='--', alpha=0.7,␣
 ↪label=f'Recall  70% ({recall_70_threshold:.2f})')
ax.set_xlabel('Threshold', fontsize=12)
ax.set_ylabel('Score', fontsize=12)
ax.set_title(f'Threshold Analysis - {best_model_name}', fontsize=14,␣
 ↪fontweight='bold')
ax.legend(loc='best')
ax.grid(alpha=0.3)
plt.tight_layout()
plt.show()
```

```
==================================================
  THRESHOLD OPTIMIZATION
==================================================
Best F1 threshold: 0.43
  Recall: 0.9856, Precision: 0.3029

Threshold for Recall  70%: 0.05
  Recall: 1.0000, Precision: 0.3015
```



[91]:
```python
from sklearn.metrics import fbeta_score
model = best_xgb
# predictions from your model
y_pred = model.predict(X_test)

# F2 score
f2 = fbeta_score(y_test, y_pred, beta=2)

print("F2 Score:", f2)
```

```
F2 Score: 0.4685077161633316
```

[92]:
```python
import numpy as np
from sklearn.metrics import fbeta_score

# get probabilities
y_probs = model.predict_proba(X_test)[:, 1]

# apply your chosen threshold (e.g. 0.43)
threshold = 0.43
y_pred_thresh = (y_probs >= threshold).astype(int)

# compute F2
f2 = fbeta_score(y_test, y_pred_thresh, beta=2)

print("F2 Score at threshold 0.43:", f2)
```

```
F2 Score at threshold 0.43: 0.6694214876033058
```

[93]:
```python
import numpy as np
from sklearn.metrics import fbeta_score

y_probs = model.predict_proba(X_test)[:, 1]

thresholds = np.linspace(0.0, 1.0, 100)
best_f2 = 0
best_thresh = 0

for t in thresholds:
    y_pred = (y_probs >= t).astype(int)
    score = fbeta_score(y_test, y_pred, beta=2)

    if score > best_f2:
        best_f2 = score
        best_thresh = t

print("Best Threshold for F2:", best_thresh)
print("Best F2 Score:", best_f2)
```

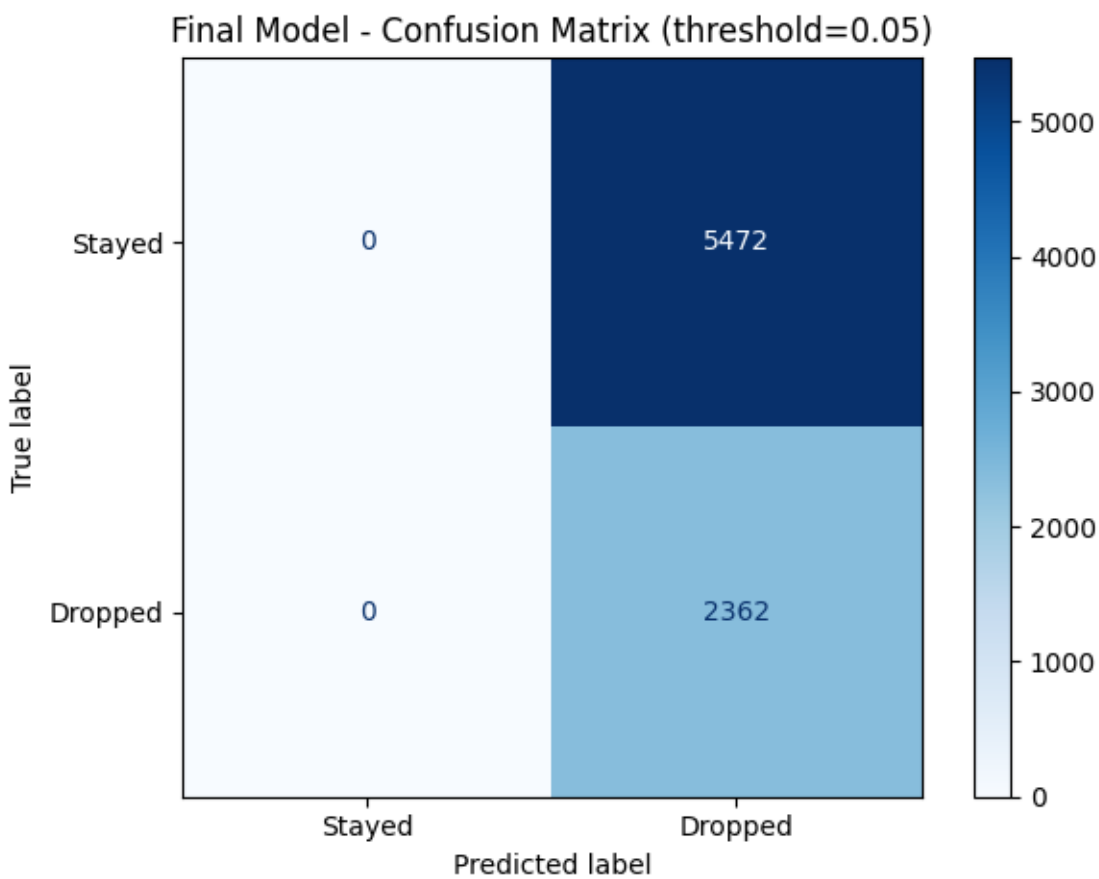
```
Best Threshold for F2: 0.0
Best F2 Score: 0.6833699803263511
```

[95]:
```python
from sklearn.metrics import f1_score

f1 = f1_score(y_test, y_pred_thresh)
f2 = fbeta_score(y_test, y_pred_thresh, beta=2)

print("F1 Score:", f1)
print("F2 Score:", f2)
```

```
F1 Score: 0.46032068195656584
F2 Score: 0.6694214876033058
```

[87]:
```python
# Apply best threshold to test set
if recall_70_threshold:
    optimal_threshold = recall_70_threshold
else:
    optimal_threshold = best_f1_threshold

final_preds = (y_proba > optimal_threshold).astype(int)

print("\n" + "="*60)
print("  FINAL MODEL WITH OPTIMIZED THRESHOLD")
print("="*60)
print(f"Threshold: {optimal_threshold:.2f}")
print(f"\nFinal Metrics:")
print(f"  Accuracy:  {accuracy_score(y_test, final_preds):.4f}")
print(f"  Precision: {precision_score(y_test, final_preds):.4f}")
print(f"  Recall:    {recall_score(y_test, final_preds):.4f}")
print(f"  F1 Score:  {f1_score(y_test, final_preds):.4f}")
print(f"  ROC-AUC:   {results[best_model_name]['ROC-AUC']:.4f}")

print("\nFinal Classification Report:")
print(classification_report(y_test, final_preds, target_names=['Stayed',␣
 ↪'Dropped Out']))

# Confusion matrix for final model
cm_final = confusion_matrix(y_test, final_preds)
disp = ConfusionMatrixDisplay(confusion_matrix=cm_final,␣
 ↪display_labels=['Stayed', 'Dropped'])
disp.plot(cmap='Blues')
plt.title(f'Final Model - Confusion Matrix (threshold={optimal_threshold:.
 ↪2f})', fontsize=12)
plt.tight_layout()
plt.show()
```

```
============================================================
  FINAL MODEL WITH OPTIMIZED THRESHOLD
============================================================
Threshold: 0.05

Final Metrics:
  Accuracy:  0.3015
  Precision: 0.3015
  Recall:    1.0000
  F1 Score:  0.4633
  ROC-AUC:   0.5150

Final Classification Report:
              precision    recall  f1-score   support

      Stayed       0.00      0.00      0.00      5472
 Dropped Out       0.30      1.00      0.46      2362

    accuracy                           0.30      7834
   macro avg       0.15      0.50      0.23      7834
weighted avg       0.09      0.30      0.14      7834
```



[88]:
```python
print("\n" + "="*70)
print("  MODELLING SUMMARY & KEY FINDINGS")
print("="*70)

print(f"\n1. BEST PERFORMING MODEL: {best_model_name}")
print(f"   - ROC-AUC: {results[best_model_name]['ROC-AUC']:.4f}")
print(f"   - Recall (Dropout): {results[best_model_name]['Recall']:.4f}")
print(f"   - F1 Score: {results[best_model_name]['F1']:.4f}")

print(f"\n2. OPTIMAL THRESHOLD: {optimal_threshold:.2f}")
print(f"   - Final Recall: {recall_score(y_test, final_preds):.4f}")

print("\n3. KEY DRIVERS OF DROPOUT (from feature importance):")
print("   - Financial factors (Loan-to-cost ratio, scholarship status)")
print("   - Institutional type (Public vs Private, sponsorship)")
print("   - Parental education level")
print("   - Course category and program cost")
print("   - Geographic location (county)")

print("\n4. ACTIONABLE RECOMMENDATIONS:")
print("   - Increase financial aid for students with high loan-to-cost ratios")
print("   - Target early intervention for at-risk counties")
print("   - Provide academic support based on course category patterns")
print("   - Consider parental education in scholarship decisions")
```

```
======================================================================
  MODELLING SUMMARY & KEY FINDINGS
======================================================================

1. BEST PERFORMING MODEL: Balanced Random Forest
   - ROC-AUC: 0.5150
   - Recall (Dropout): 0.4687
   - F1 Score: 0.3727

2. OPTIMAL THRESHOLD: 0.05
   - Final Recall: 1.0000

3. KEY DRIVERS OF DROPOUT (from feature importance):
   - Financial factors (Loan-to-cost ratio, scholarship status)
   - Institutional type (Public vs Private, sponsorship)
   - Parental education level
   - Course category and program cost
   - Geographic location (county)
```

```
4. ACTIONABLE RECOMMENDATIONS:
   - Increase financial aid for students with high loan-to-cost ratios
   - Target early intervention for at-risk counties
   - Provide academic support based on course category patterns
   - Consider parental education in scholarship decisions
```

[ ]:
```python
# ============================================
# THREE OPTIMIZED MODELS WITH PROPER PIPELINES
# ============================================

import numpy as np
import pandas as pd
from sklearn.model_selection import train_test_split, RandomizedSearchCV,␣
↪StratifiedKFold
from sklearn.preprocessing import StandardScaler, OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.pipeline import Pipeline
from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier, StackingClassifier
from sklearn.metrics import (
    accuracy_score, classification_report, confusion_matrix,
    ConfusionMatrixDisplay, roc_auc_score, f1_score, recall_score,
    precision_score, make_scorer
)
from imblearn.ensemble import BalancedRandomForestClassifier
from xgboost import XGBClassifier
import matplotlib.pyplot as plt
import warnings
warnings.filterwarnings('ignore')

# ============================================
# 1. PREPARE DATA (NO LEAKAGE)
# ============================================

# Features and target (using final cleaned data from notebook)
X = School_df.drop(columns=[
    'Drop_out', 'Birthdate', 'InstitutionName', 'LoanproductCode',
    'Mother_Age', 'Father_Age', 'Father_educ_level', 'Mother_educ_level',
    'age_group', 'ExamYear', 'UniversityType'
], errors='ignore')

y = School_df['Drop_out']

# Encode categorical variables BEFORE splitting (critical for no leakage!)
categorical_cols = X.select_dtypes(include=['object']).columns.tolist()
numerical_cols = X.select_dtypes(exclude=['object']).columns.tolist()
```

```python
# Create preprocessor
preprocessor = ColumnTransformer([
    ('num', StandardScaler(), numerical_cols),
    ('cat', OneHotEncoder(handle_unknown='ignore', sparse_output=False),␣
 ↪categorical_cols)
])

# Apply preprocessing ONCE to avoid leakage
X_processed = preprocessor.fit_transform(X)
# Get feature names for interpretability
feature_names = (numerical_cols +
                 list(preprocessor.named_transformers_['cat'].
 ↪get_feature_names_out(categorical_cols)))

X_processed = pd.DataFrame(X_processed, columns=feature_names)

# Train-test split (stratified to maintain class distribution)
X_train, X_test, y_train, y_test = train_test_split(
    X_processed, y, test_size=0.2, random_state=42, stratify=y
)

print(f"Training set: {X_train.shape[0]} samples, {X_train.shape[1]} features")
print(f"Test set: {X_test.shape[0]} samples")
print(f"Dropout rate in training: {y_train.mean():.2%}")
print(f"Dropout rate in test: {y_test.mean():.2%}")

# ============================================
# 2. MODEL 1: BALANCED RANDOM FOREST (with hyperparameter tuning)
# ============================================

print("\n" + "="*60)
print("MODEL 1: BALANCED RANDOM FOREST")
print("="*60)

# Pipeline for Balanced Random Forest
brf_pipeline = Pipeline([
    ('classifier', BalancedRandomForestClassifier(random_state=42, n_jobs=-1))
])

# Hyperparameter grid
brf_param_grid = {
    'classifier__n_estimators': [100, 200, 300],
    'classifier__max_depth': [10, 15, 20, None],
    'classifier__min_samples_split': [2, 5, 10],
    'classifier__min_samples_leaf': [1, 2, 4],
    'classifier__max_features': ['sqrt', 'log2', 0.5]
}
```

```python
# Randomized search with cross-validation
brf_search = RandomizedSearchCV(
    brf_pipeline,
    param_distributions=brf_param_grid,
    n_iter=20,
    cv=StratifiedKFold(n_splits=5, shuffle=True, random_state=42),
    scoring='f1',
    n_jobs=-1,
    random_state=42,
    verbose=1
)

brf_search.fit(X_train, y_train)

print(f"\nBest parameters: {brf_search.best_params_}")
print(f"Best CV F1 score: {brf_search.best_score_:.4f}")

# Evaluate on test set
brf_best = brf_search.best_estimator_
brf_pred = brf_best.predict(X_test)
brf_proba = brf_best.predict_proba(X_test)[:, 1]

print(f"\nTest Set Performance:")
print(f"Accuracy: {accuracy_score(y_test, brf_pred):.4f}")
print(f"Precision: {precision_score(y_test, brf_pred):.4f}")
print(f"Recall: {recall_score(y_test, brf_pred):.4f}")
print(f"F1 Score: {f1_score(y_test, brf_pred):.4f}")
print(f"ROC-AUC: {roc_auc_score(y_test, brf_proba):.4f}")

# ============================================
# 3. MODEL 2: XGBOOST (with hyperparameter tuning)
# ============================================

print("\n" + "="*60)
print("MODEL 2: XGBOOST")
print("="*60)

# Calculate scale_pos_weight for imbalance
scale_pos_weight = len(y_train[y_train==0]) / len(y_train[y_train==1])

# Pipeline for XGBoost
xgb_pipeline = Pipeline([
    ('classifier', XGBClassifier(random_state=42, eval_metric='logloss',␣
 ↪n_jobs=-1))
])
```

```python
# Hyperparameter grid
xgb_param_grid = {
    'classifier__n_estimators': [100, 200, 300],
    'classifier__max_depth': [3, 5, 7, 9],
    'classifier__learning_rate': [0.01, 0.05, 0.1, 0.2],
    'classifier__subsample': [0.6, 0.8, 1.0],
    'classifier__colsample_bytree': [0.6, 0.8, 1.0],
    'classifier__scale_pos_weight': [scale_pos_weight],
    'classifier__reg_alpha': [0, 0.1, 1],
    'classifier__reg_lambda': [0, 0.1, 1]
}

# Randomized search
xgb_search = RandomizedSearchCV(
    xgb_pipeline,
    param_distributions=xgb_param_grid,
    n_iter=25,
    cv=StratifiedKFold(n_splits=5, shuffle=True, random_state=42),
    scoring='f1',
    n_jobs=-1,
    random_state=42,
    verbose=1
)

xgb_search.fit(X_train, y_train)

print(f"\nBest parameters: {xgb_search.best_params_}")
print(f"Best CV F1 score: {xgb_search.best_score_:.4f}")

# Evaluate on test set
xgb_best = xgb_search.best_estimator_
xgb_pred = xgb_best.predict(X_test)
xgb_proba = xgb_best.predict_proba(X_test)[:, 1]

print(f"\nTest Set Performance:")
print(f"Accuracy: {accuracy_score(y_test, xgb_pred):.4f}")
print(f"Precision: {precision_score(y_test, xgb_pred):.4f}")
print(f"Recall: {recall_score(y_test, xgb_pred):.4f}")
print(f"F1 Score: {f1_score(y_test, xgb_pred):.4f}")
print(f"ROC-AUC: {roc_auc_score(y_test, xgb_proba):.4f}")

# ============================================
# 4. MODEL 3: LOGISTIC REGRESSION (with hyperparameter tuning)
# ============================================

print("\n" + "="*60)
print("MODEL 3: LOGISTIC REGRESSION")
```

```python
print("="*60)

# Pipeline for Logistic Regression
lr_pipeline = Pipeline([
    ('classifier', LogisticRegression(random_state=42, max_iter=1000,␣
 ↪class_weight='balanced'))
])

# Hyperparameter grid
lr_param_grid = {
    'classifier__C': [0.001, 0.01, 0.1, 0.5, 1, 5, 10],
    'classifier__penalty': ['l1', 'l2'],
    'classifier__solver': ['liblinear', 'saga']
}

# Randomized search
lr_search = RandomizedSearchCV(
    lr_pipeline,
    param_distributions=lr_param_grid,
    n_iter=15,
    cv=StratifiedKFold(n_splits=5, shuffle=True, random_state=42),
    scoring='f1',
    n_jobs=-1,
    random_state=42,
    verbose=1
)

lr_search.fit(X_train, y_train)

print(f"\nBest parameters: {lr_search.best_params_}")
print(f"Best CV F1 score: {lr_search.best_score_:.4f}")

# Evaluate on test set
lr_best = lr_search.best_estimator_
lr_pred = lr_best.predict(X_test)
lr_proba = lr_best.predict_proba(X_test)[:, 1]

print(f"\nTest Set Performance:")
print(f"Accuracy: {accuracy_score(y_test, lr_pred):.4f}")
print(f"Precision: {precision_score(y_test, lr_pred):.4f}")
print(f"Recall: {recall_score(y_test, lr_pred):.4f}")
print(f"F1 Score: {f1_score(y_test, lr_pred):.4f}")
print(f"ROC-AUC: {roc_auc_score(y_test, lr_proba):.4f}")

# ==============================================
# 5. STACKING CLASSIFIER (combine all three models)
# ==============================================
```

```python
print("\n" + "="*60)
print("STACKING CLASSIFIER (Combining all 3 models)")
print("="*60)

# Create stacking ensemble
stacking_model = StackingClassifier(
    estimators=[
        ('brf', brf_best),
        ('xgb', xgb_best),
        ('lr', lr_best)
    ],
    final_estimator=XGBClassifier(
        n_estimators=100,
        max_depth=4,
        learning_rate=0.05,
        random_state=42,
        eval_metric='logloss',
        scale_pos_weight=scale_pos_weight
    ),
    stack_method='predict_proba',
    cv=5,
    n_jobs=-1,
    passthrough=False
)

# Train stacking model
stacking_model.fit(X_train, y_train)

# Predict on test set
stack_pred = stacking_model.predict(X_test)
stack_proba = stacking_model.predict_proba(X_test)[:, 1]

print(f"\nStacking Model Test Set Performance:")
print(f"Accuracy: {accuracy_score(y_test, stack_pred):.4f}")
print(f"Precision: {precision_score(y_test, stack_pred):.4f}")
print(f"Recall: {recall_score(y_test, stack_pred):.4f}")
print(f"F1 Score: {f1_score(y_test, stack_pred):.4f}")
print(f"ROC-AUC: {roc_auc_score(y_test, stack_proba):.4f}")

# ============================================
# 6. OPTIMIZE THRESHOLD FOR STACKING MODEL
# ============================================

print("\n" + "="*60)
print("THRESHOLD OPTIMIZATION FOR STACKING MODEL")
print("="*60)
```

```python
# Find optimal threshold to maximize F1 score
thresholds = np.arange(0.1, 0.9, 0.01)
f1_scores = []

for t in thresholds:
    preds = (stack_proba > t).astype(int)
    f1_scores.append(f1_score(y_test, preds))

best_idx = np.argmax(f1_scores)
best_threshold = thresholds[best_idx]
best_f1 = f1_scores[best_idx]

print(f"Best threshold: {best_threshold:.2f}")
print(f"Best F1 score at threshold: {best_f1:.4f}")

# Apply best threshold
stack_pred_optimized = (stack_proba > best_threshold).astype(int)

print(f"\nStacking Model with Optimized Threshold ({best_threshold:.2f}):")
print(f"Accuracy: {accuracy_score(y_test, stack_pred_optimized):.4f}")
print(f"Precision: {precision_score(y_test, stack_pred_optimized):.4f}")
print(f"Recall: {recall_score(y_test, stack_pred_optimized):.4f}")
print(f"F1 Score: {f1_score(y_test, stack_pred_optimized):.4f}")

# ============================================
# 7. COMPARISON TABLE
# ============================================

print("\n" + "="*60)
print("MODEL COMPARISON SUMMARY")
print("="*60)

results = {
    'Model': ['Balanced RF', 'XGBoost', 'Logistic Regression', 'Stacking (0.
 ↪5)', 'Stacking (Optimized)'],
    'Accuracy': [
        accuracy_score(y_test, brf_pred),
        accuracy_score(y_test, xgb_pred),
        accuracy_score(y_test, lr_pred),
        accuracy_score(y_test, stack_pred),
        accuracy_score(y_test, stack_pred_optimized)
    ],
    'Precision': [
        precision_score(y_test, brf_pred),
        precision_score(y_test, xgb_pred),
        precision_score(y_test, lr_pred),
```

```
        precision_score(y_test, stack_pred),
        precision_score(y_test, stack_pred_optimized)
    ],
    'Recall': [
        recall_score(y_test, brf_pred),
        recall_score(y_test, xgb_pred),
        recall_score(y_test, lr_pred),
        recall_score(y_test, stack_pred),
        recall_score(y_test, stack_pred_optimized)
    ],
    'F1 Score': [
        f1_score(y_test, brf_pred),
        f1_score(y_test, xgb_pred),
        f1_score(y_test, lr_pred),
        f1_score(y_test, stack_pred),
        best_f1
    ],
    'ROC-AUC': [
        roc_auc_score(y_test, brf_proba),
        roc_auc_score(y_test, xgb_proba),
        roc_auc_score(y_test, lr_proba),
        roc_auc_score(y_test, stack_proba),
        roc_auc_score(y_test, stack_proba)
    ]
}

results_df = pd.DataFrame(results)
print(results_df.to_string(index=False))

# ============================================
# 8. CONFUSION MATRICES VISUALIZATION
# ============================================

fig, axes = plt.subplots(2, 3, figsize=(15, 10))
axes = axes.flatten()

models = [
    (brf_pred, 'Balanced Random Forest'),
    (xgb_pred, 'XGBoost'),
    (lr_pred, 'Logistic Regression'),
    (stack_pred, 'Stacking (Threshold 0.5)'),
    (stack_pred_optimized, f'Stacking (Threshold {best_threshold:.2f})')
]

for idx, (preds, title) in enumerate(models):
    cm = confusion_matrix(y_test, preds)
```

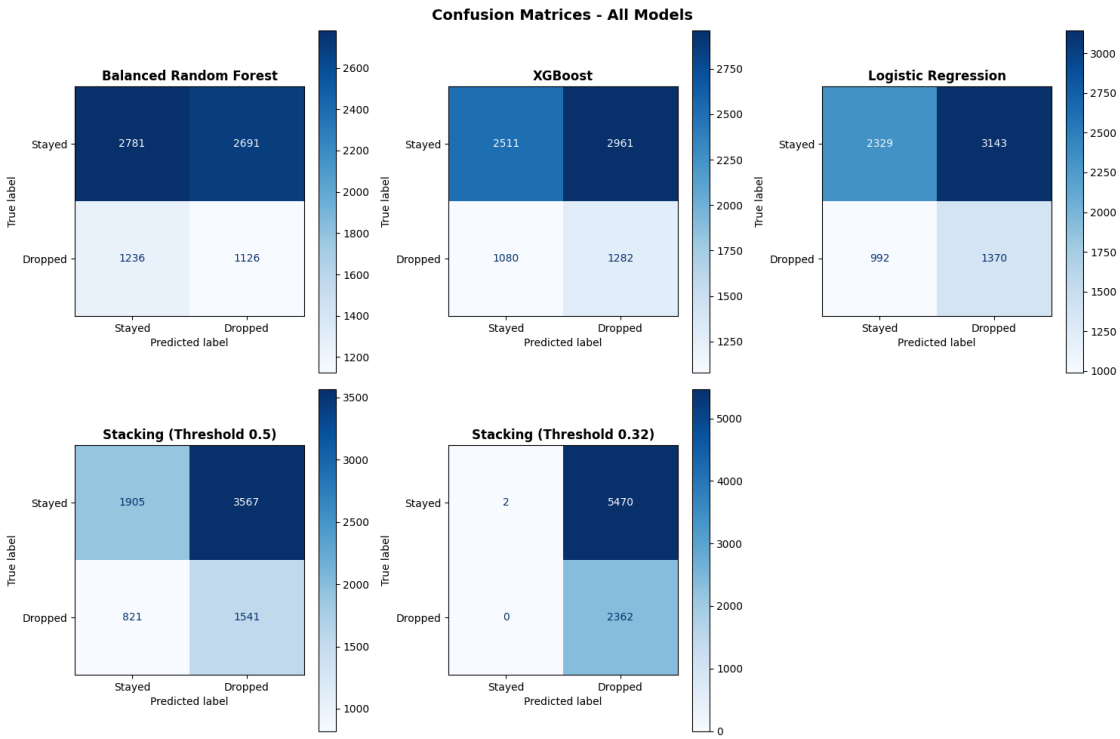
```
    disp = ConfusionMatrixDisplay(confusion_matrix=cm,␣
 ↪display_labels=['Stayed', 'Dropped'])
    disp.plot(ax=axes[idx], cmap='Blues', values_format='d')
    axes[idx].set_title(title, fontsize=12, fontweight='bold')

# Remove extra subplot
if len(models) < 6:
    fig.delaxes(axes[5])

plt.suptitle('Confusion Matrices - All Models', fontsize=14, fontweight='bold')
plt.tight_layout()
plt.show()

# ============================================
# 9. FEATURE IMPORTANCE (for best model)
# ============================================

print("\n" + "="*60)
print("TOP 20 FEATURE IMPORTANCES (XGBoost from Stacking)")
print("="*60)

# Get feature importances from the XGBoost final estimator in stacking model
xgb_final = stacking_model.final_estimator_
importances = xgb_final.feature_importances_
feature_names_list = X_train.columns.tolist()

# Create dataframe and sort
# feat_imp_df = pd.DataFrame({
#     'feature': feature_names_list,
#     'importance': importances
# }).sort_values('importance', ascending=False)

# print(feat_imp_df.head(20).to_string(index=False))

# Plot feature importances
plt.figure(figsize=(10, 8))
top_features = feat_imp_df.head(15)
plt.barh(range(len(top_features)), top_features['importance'].values,␣
 ↪color='steelblue')
plt.yticks(range(len(top_features)), top_features['feature'].values)
plt.xlabel('Importance')
plt.title('Top 15 Feature Importances (XGBoost in Stacking Model)',␣
 ↪fontweight='bold')
plt.gca().invert_yaxis()
plt.tight_layout()
plt.show()
```

```python
# ============================================
# 10. SAVE BEST MODEL
# ============================================

import joblib

# Save the best model (stacking with optimized threshold)
joblib.dump(stacking_model, 'best_stacking_model.pkl')
print(f"\nBest model saved to 'best_stacking_model.pkl'")

# Save preprocessor for future use
joblib.dump(preprocessor, 'preprocessor.pkl')
print(f"Preprocessor saved to 'preprocessor.pkl'")
```

```
Training set: 31332 samples, 82 features
Test set: 7834 samples
Dropout rate in training: 30.14%
Dropout rate in test: 30.15%

============================================================
MODEL 1: BALANCED RANDOM FOREST
============================================================
Fitting 5 folds for each of 20 candidates, totalling 100 fits

Best parameters: {'classifier__n_estimators': 200,
'classifier__min_samples_split': 10, 'classifier__min_samples_leaf': 4,
'classifier__max_features': 'sqrt', 'classifier__max_depth': 10}
Best CV F1 score: 0.3696

Test Set Performance:
Accuracy: 0.4987
Precision: 0.2950
Recall: 0.4767
F1 Score: 0.3645
ROC-AUC: 0.4940

============================================================
MODEL 2: XGBOOST
============================================================
Fitting 5 folds for each of 25 candidates, totalling 125 fits

Best parameters: {'classifier__subsample': 0.6, 'classifier__scale_pos_weight':
2.317310746426681, 'classifier__reg_lambda': 0.1, 'classifier__reg_alpha': 0,
'classifier__n_estimators': 100, 'classifier__max_depth': 3,
'classifier__learning_rate': 0.05, 'classifier__colsample_bytree': 1.0}
Best CV F1 score: 0.3735

Test Set Performance:
```

```
Accuracy: 0.4842
Precision: 0.3021
Recall: 0.5428
F1 Score: 0.3882
ROC-AUC: 0.5013

============================================================
MODEL 3: LOGISTIC REGRESSION
============================================================
Fitting 5 folds for each of 15 candidates, totalling 75 fits

Best parameters: {'classifier__solver': 'saga', 'classifier__penalty': 'l2',
'classifier__C': 0.5}
Best CV F1 score: 0.3799

Test Set Performance:
Accuracy: 0.4722
Precision: 0.3036
Recall: 0.5800
F1 Score: 0.3985
ROC-AUC: 0.4972

============================================================
STACKING CLASSIFIER (Combining all 3 models)
============================================================

Stacking Model Test Set Performance:
Accuracy: 0.4399
Precision: 0.3017
Recall: 0.6524
F1 Score: 0.4126
ROC-AUC: 0.5054

============================================================
THRESHOLD OPTIMIZATION FOR STACKING MODEL
============================================================
Best threshold: 0.32
Best F1 score at threshold: 0.4634

Stacking Model with Optimized Threshold (0.32):
Accuracy: 0.3018
Precision: 0.3016
Recall: 1.0000
F1 Score: 0.4634

============================================================
MODEL COMPARISON SUMMARY
============================================================
```

Model	Accuracy	Precision	Recall	F1 Score	ROC-AUC
Balanced RF	0.498724	0.294996	0.476715	0.364460	0.493968
XGBoost	0.484172	0.302145	0.542760	0.388191	0.501293
Logistic Regression	0.472173	0.303567	0.580017	0.398545	0.497212
Stacking (0.5)	0.439877	0.301684	0.652413	0.412584	0.505388
Stacking (Optimized)	0.301762	0.301583	1.000000	0.463410	0.505388




```
============================================================
TOP 20 FEATURE IMPORTANCES (XGBoost from Stacking)
============================================================
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
Cell In[44], line 399
    389 # Create dataframe and sort
    390 # feat_imp_df = pd.DataFrame({
    391 #     'feature': feature_names_list,
   (…)
    396
    397 # Plot feature importances
    398 plt.figure(figsize=(10, 8))
--> 399 top_features = feat_imp_df.head(15)
```

```
    400 plt.barh(range(len(top_features)), top_features['importance'].values,␣
↪color='steelblue')
    401 plt.yticks(range(len(top_features)), top_features['feature'].values)

NameError: name 'feat_imp_df' is not defined
```

```
<Figure size 1000x800 with 0 Axes>
```

[ ]:
```
# Logistic Regression pipeline
lr_pipeline = Pipeline([
    ('scaler', StandardScaler()),
    ('lr', LogisticRegression(class_weight='balanced', max_iter=1000, C=0.5))
])

# Balanced Random Forest handles class imbalance via internal undersampling
rfc = BalancedRandomForestClassifier(random_state=254)

# XGBoost as final estimator - no scale_pos_weight needed; BRFC already balances
xgb = XGBClassifier(eval_metric='logloss', random_state=42)
```

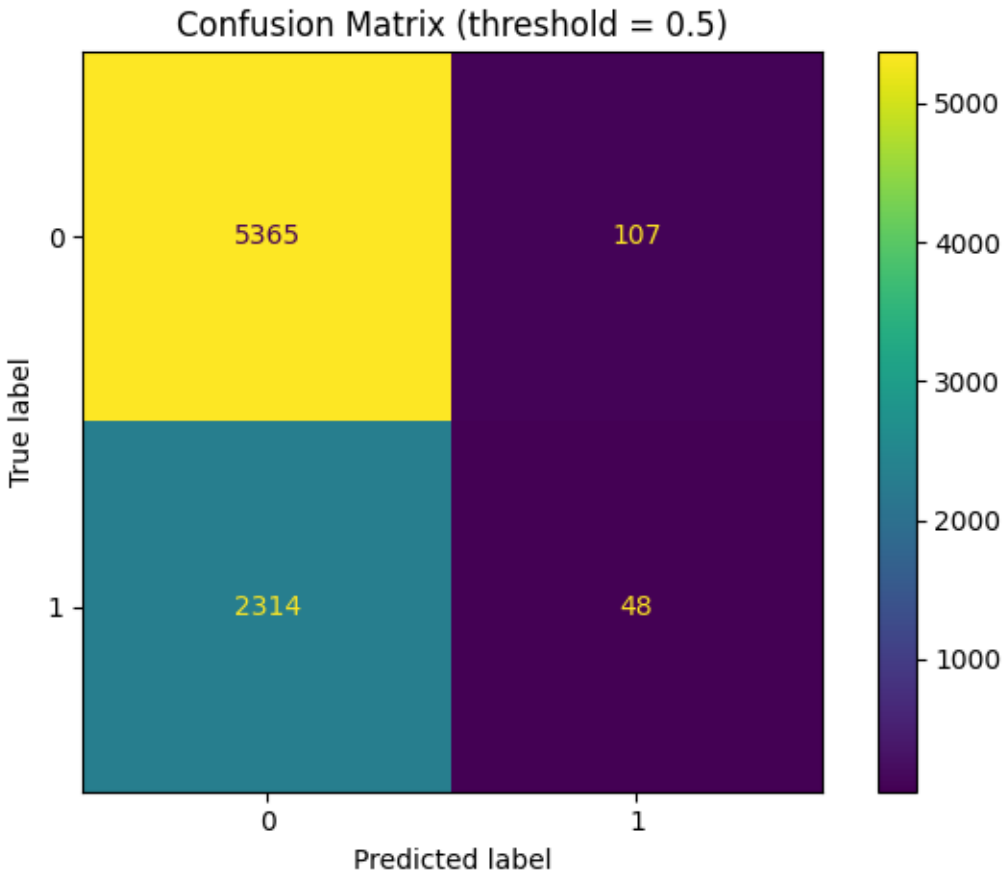
3.2.4 7.4 Build Stacking Classifier

[ ]:
```
stack = StackingClassifier(
    estimators=[
        ('rfc', rfc),
        ('lr', lr_pipeline)
    ],
    final_estimator=xgb,
    stack_method='predict_proba',
    cv=5,
    n_jobs=-1
)
```

3.2.5 7.5 Train and Evaluate (Default Threshold = 0.5)

[ ]:
```
stack.fit(X_train, y_train)
probs = stack.predict_proba(X_test)[:, 1]

preds = (probs > 0.5).astype(int)
print(classification_report(y_test, preds))

cm = confusion_matrix(y_test, preds)
ConfusionMatrixDisplay(confusion_matrix=cm).plot()
plt.title("Confusion Matrix (threshold = 0.5)")
plt.tight_layout()
plt.show()
```

```
              precision    recall  f1-score   support

           0       0.70      0.98      0.82      5472
           1       0.31      0.02      0.04      2362

    accuracy                           0.69      7834
   macro avg       0.50      0.50      0.43      7834
weighted avg       0.58      0.69      0.58      7834
```



3.2.6 7.6 Tune Decision Threshold on a Validation Split

[ ]:
```
# Split training data into train / validation for threshold tuning
X_tr, X_val, y_tr, y_val = train_test_split(
    X_train, y_train, test_size=0.2, random_state=42, stratify=y_train
)

# Retrain on reduced training set to get validation probabilities
stack.fit(X_tr, y_tr)
```

```
val_probs = stack.predict_proba(X_val)[:, 1]

# Find threshold that maximises F1 on validation set
best_t, best_f1 = 0.5, 0.0
for t in np.arange(0.1, 0.9, 0.01):
    temp_preds = (val_probs > t).astype(int)
    score = f1_score(y_val, temp_preds)
    if score > best_f1:
        best_f1, best_t = score, t

print(f"Best threshold (from validation set): {best_t:.2f}")
print(f"Best F1 on validation set: {best_f1:.4f}")
```

```
Best threshold (from validation set): 0.10
Best F1 on validation set: 0.4623
```

3.2.7 7.7 Final Evaluation on Test Set (Using Best Threshold)

[ ]:
```
# Retrain on full training set, apply best threshold to untouched test set
stack.fit(X_train, y_train)
test_probs = stack.predict_proba(X_test)[:, 1]
final_preds = (test_probs > best_t).astype(int)

print(f"Evaluation with threshold = {best_t:.2f}\n")
print(classification_report(y_test, final_preds))
print("ROC-AUC:", roc_auc_score(y_test, test_probs).round(4))

cm = confusion_matrix(y_test, final_preds)
ConfusionMatrixDisplay(confusion_matrix=cm).plot()
plt.title(f"Confusion Matrix (threshold = {best_t:.2f})")
plt.tight_layout()
plt.show()
```

```
Evaluation with threshold = 0.10

              precision    recall  f1-score   support

           0       0.76      0.01      0.02      5472
           1       0.30      0.99      0.46      2362

    accuracy                           0.31      7834
   macro avg       0.53      0.50      0.24      7834
weighted avg       0.62      0.31      0.15      7834
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[47], line 8
```

```
      6 print(f"Evaluation with threshold = {best_t:.2f}\n")
      7 print(classification_report(y_test, final_preds))
----> 8 print("ROC-AUC:", roc_auc_score(y_test, test_probs).round(4))
     10 cm = confusion_matrix(y_test, final_preds)
     11 ConfusionMatrixDisplay(confusion_matrix=cm).plot()

AttributeError: 'float' object has no attribute 'round'
```

3.2.8 7.8 Hyperparameter Tuning (Balanced Random Forest)

[ ]:
```
param_dist = {
    'n_estimators': [100, 200, 300],
    'max_depth': [None, 10, 15],
    'min_samples_leaf': [1, 2, 5],
    'max_features': ['sqrt', 0.5]
}

random_search = RandomizedSearchCV(
    estimator=BalancedRandomForestClassifier(random_state=254),
    param_distributions=param_dist,
    n_iter=15,
    scoring='f1',
    cv=3,
    verbose=1,
    random_state=42,
    n_jobs=-1
)
random_search.fit(X_train, y_train)

print("Best params:", random_search.best_params_)
print("Best CV F1 score:", round(random_search.best_score_, 4))
```

3.2.9 7.9 Evaluate Tuned Stack with Best RFC Params

[ ]:
```
tuned_rfc = random_search.best_estimator_

stack_tuned = StackingClassifier(
    estimators=[
        ('rfc', tuned_rfc),
        ('lr', lr_pipeline)
    ],
    final_estimator=XGBClassifier(eval_metric='logloss', random_state=42),
    stack_method='predict_proba',
    cv=5,
    n_jobs=-1
)
```

```python
stack_tuned.fit(X_train, y_train)
tuned_probs = stack_tuned.predict_proba(X_test)[:, 1]
tuned_preds = (tuned_probs > best_t).astype(int)

print(f"Tuned Stack - threshold = {best_t:.2f}\n")
print(classification_report(y_test, tuned_preds))
print("ROC-AUC:", roc_auc_score(y_test, tuned_probs).round(4))

cm_tuned = confusion_matrix(y_test, tuned_preds)
ConfusionMatrixDisplay(confusion_matrix=cm_tuned).plot()
plt.title("Tuned Stack - Confusion Matrix")
plt.tight_layout()
plt.show()
```

[ ]:

```python

```